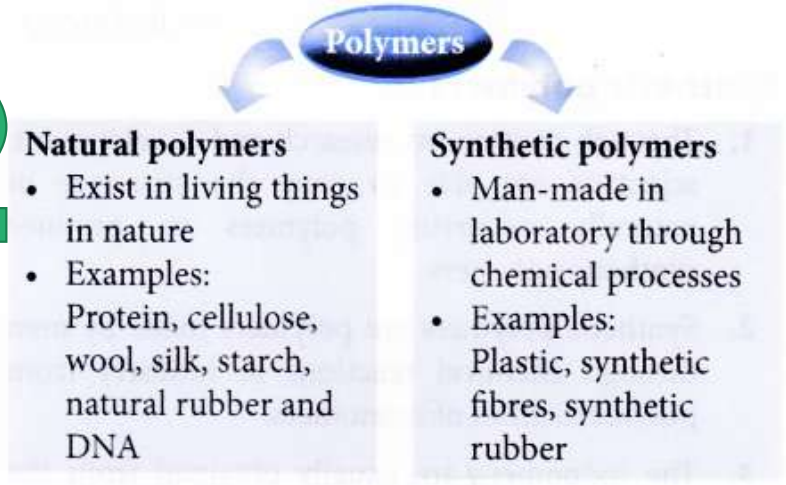
Biomedical Materials:
Polymeric biomaterials
Asst. Prof. Dr. Sumama

Introduction

WHAT IS A POLYMER?

A polymer is any of a class of natural or synthetic substances composed of very large molecules, called macromolecules, which are multiples of simpler chemical units called monomers.

Classification of Polymers based on source

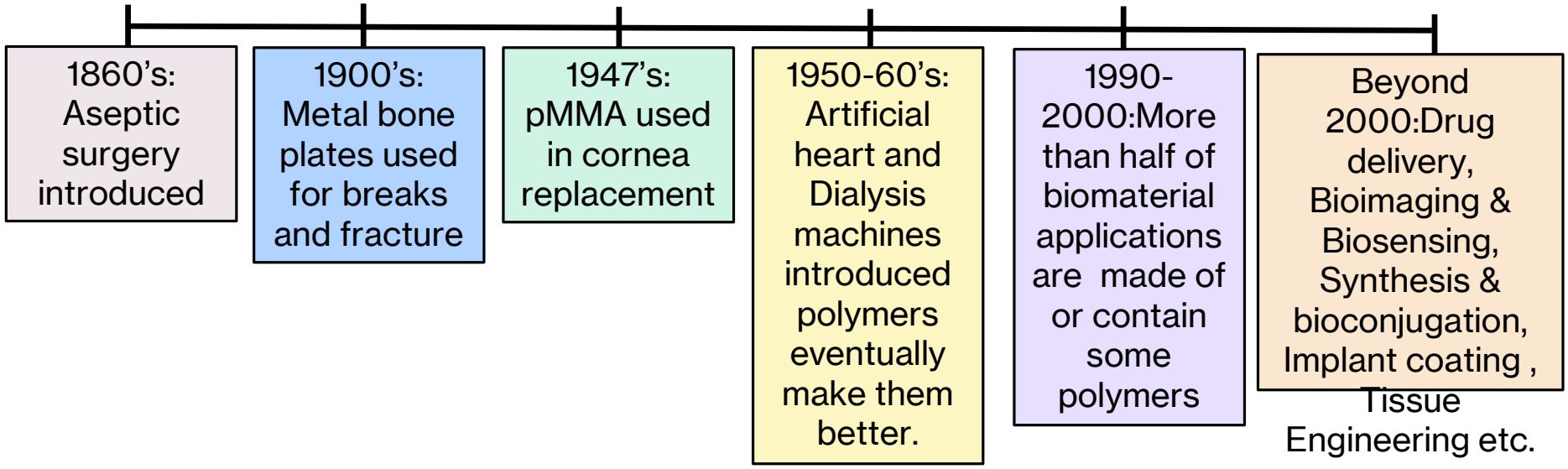


Types of Polymers

- Polymers in our body
  - DNA
  - RNA
  - proteins
  - polysaccharides
- Polymers in Nature
  - cellulose (cotton, paper)
  - starch (corn, rice)
  - natural rubber
  - silicates and ceramics
- Synthetic polymers
  - Nylon
  - polyester
  - Kevlar
  - Teflon
  - polyethylene
  - polyurethane
  - polycarbonate
  - polyimide
  - .....

HISTORY

A Brief Timeline of Polymers in Medicine



WHY POLYMERS ARE POPULAR?



Inexpensive and easy to fabricate

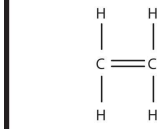
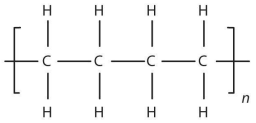
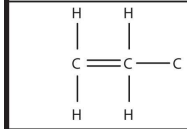
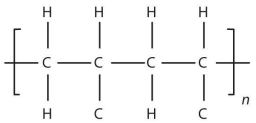
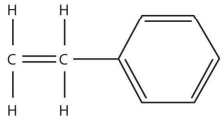
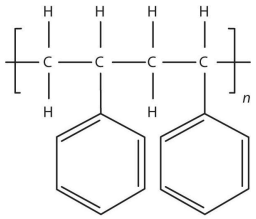
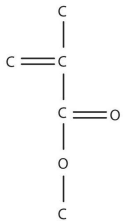
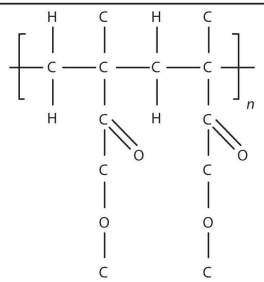
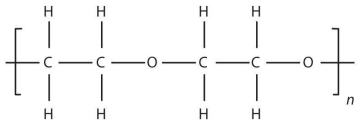
Light and strong

Abundant and versatile

WHY ARE POLYMERS USED AS BIOMEDICAL MATERIALS?

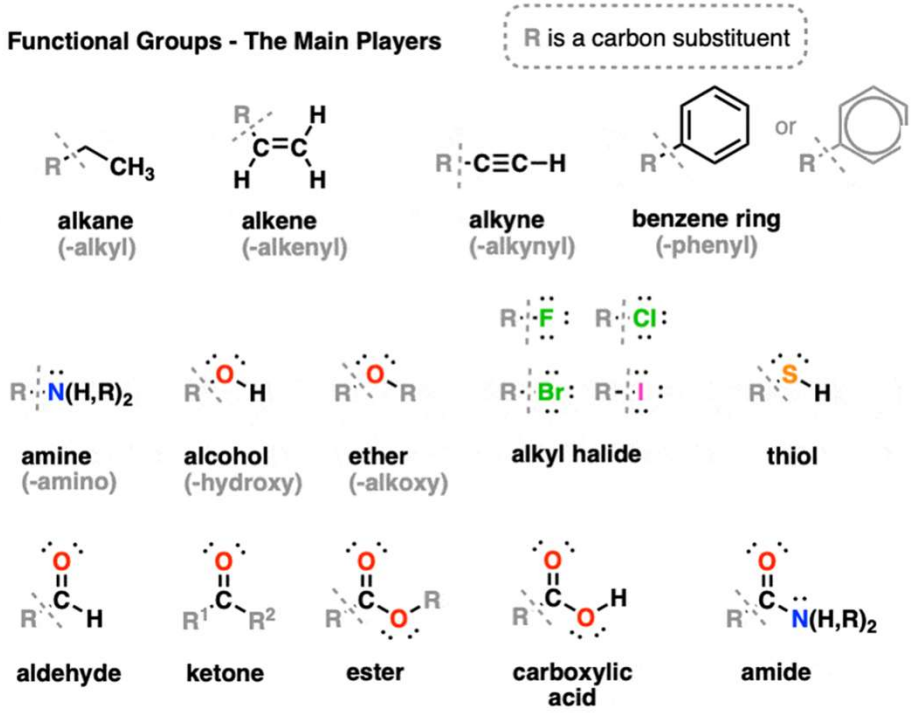
- **Biocompatibility**: Many polymers exhibit excellent compatibility with body tissues, minimizing immune response (Williams, 2009).
- **Flexibility**: Unlike metals and ceramics, polymers can be engineered to be flexible, which is beneficial for applications in soft tissues such as cardiovascular grafts and catheters.
- **Tailorability**: Polymers can be synthesized or modified to exhibit specific mechanical properties, degradation rates, and drug-releasing capabilities. For example, PLGA can be tailored for timed degradation in drug delivery (Middleton & Tipton, 2000).
- **Cost-Effectiveness**: Polymer-based devices are often less expensive to manufacture compared to metal or ceramic counterparts.
- **Sterilizability**: Most polymers can be sterilized using standard methods without significant degradation, enabling safe usage in clinical settings.

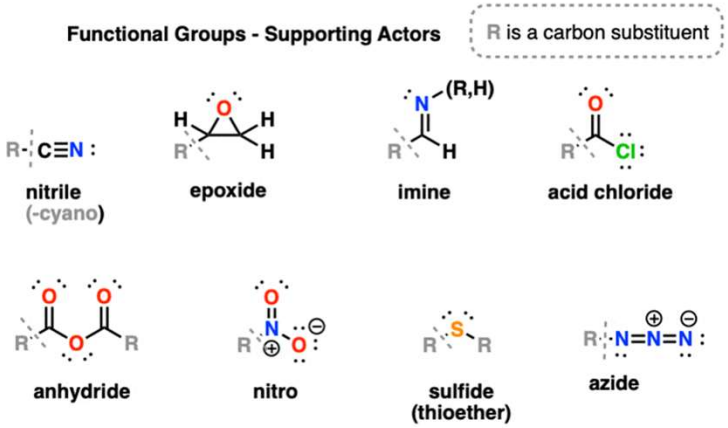
EXAMPLES OF MONOMERS & POLYMERS

Monomer	Monomer Name	Polymer	Polymer Name
	ethylene		poly(ethylene)
	propylene		poly(propylene)
	styrene		poly(styrene)
	methyl methacrylate		poly(methyl methacrylate)
	ethylene oxide		poly(ethylene oxide)

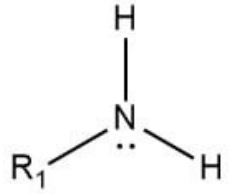
BASICS

A functional group is a specific atom or group of atoms within a molecule that determines the characteristic chemical reactions and properties of that molecule. Functional groups are important in determining a compound's reactivity and the types of chemical reactions it can participate in. When considering polymers, functional groups can also change the mechanical, chemical, optical, and electrical properties of that material.

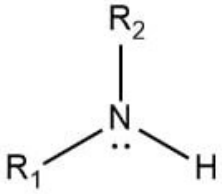




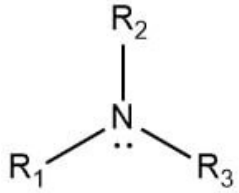
BASICS



Primary Amine



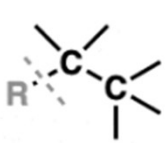
Secondary Amine



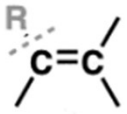
Tertiary Amine

Hydrocarbon functional groups

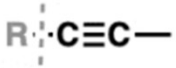
R is a carbon substituent



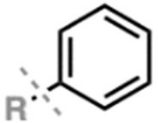
alkane
(-alkyl)



alkene
(-alkenyl)



alkyne
(-alkynyl)



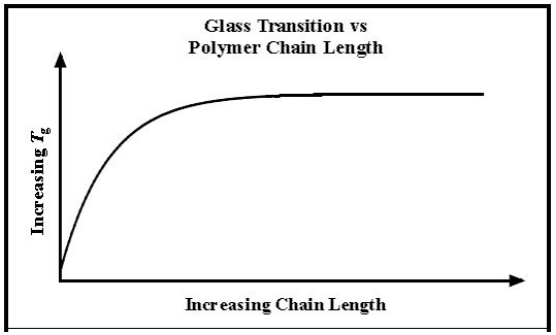
benzene ring
(-phenyl)

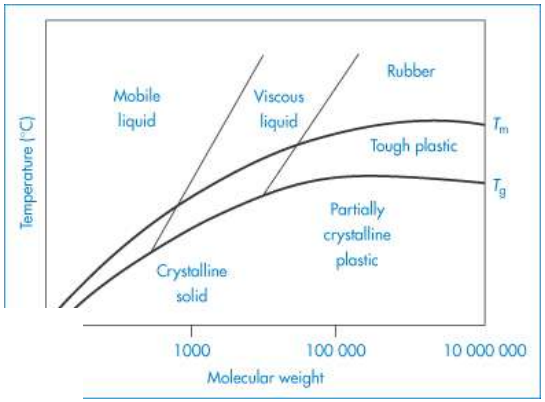
- non-polar
- extremely weak acids

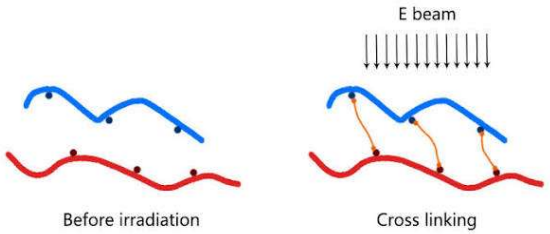
electronegativity of carbon: **2.5**
electronegativity of hydrogen: **2.2**

FACTORS AFFECTING PROPERTIES

Chain length
Crystallinity
Molecular weight
Branching
Cross-linking
Additives
Density
Chemical structure (functional groups)
Porosity
Etc.,







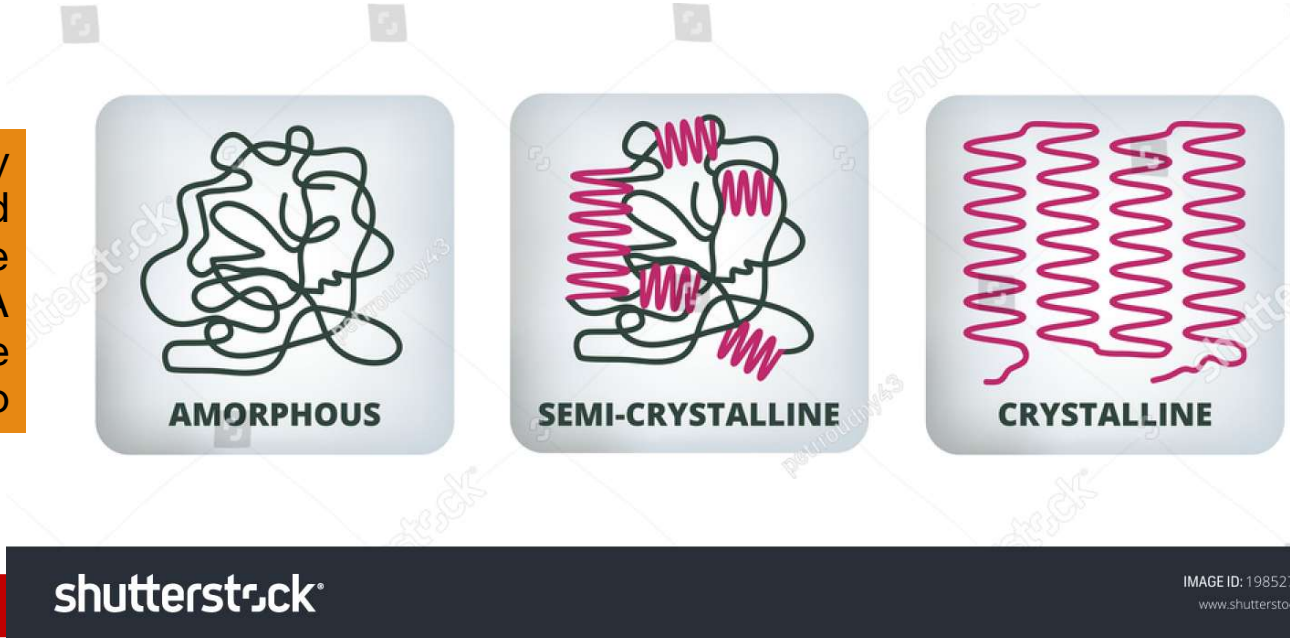
CRYSTALLİNİTY OF POLYMERS

Partiularly melting temperature and glass transition temperature are important aspects in the process of selection of polymers.

Glass transition temperature (T_g): Non-crystalline material is hard and brittle below T_g. However, above Tg, the material transforms into a rubbery substance (glass transition is observed in amorphous or semicrystalline polymers only not in crystalline ones!)

Melting: change of a solid into a liquid when heat is applied substance.

Determination of polymer crystallinity is important because mechanical and even optical properties of crystalline and amorphous polymers differ. A crystalline polymer tend to have higher melting point compared to amorphous counterpart.



NATURAL POLYMERS

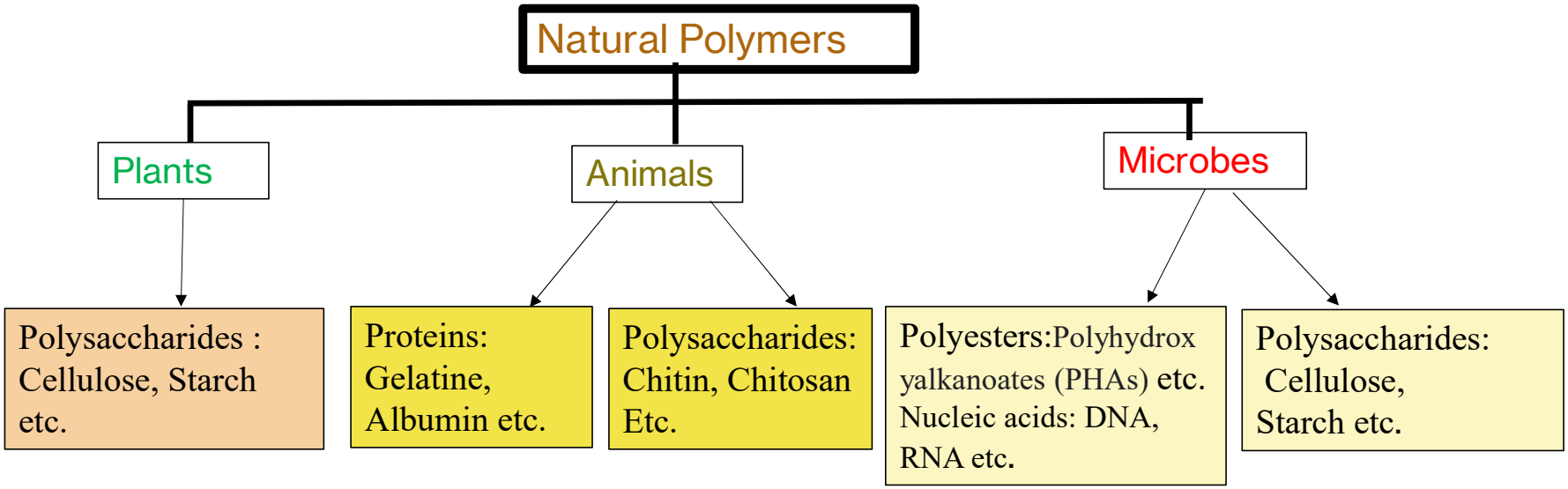
Natural polymers have been widely used in biomedical applications such as **pharmaceuticals, tissue regeneration scaffolds, drug delivery agents, and imaging agents**.

Natural polymers, also called **biopolymers**, are naturally occurring materials, formed during the life cycles of **green plants, animals, bacteria, and fungi**.

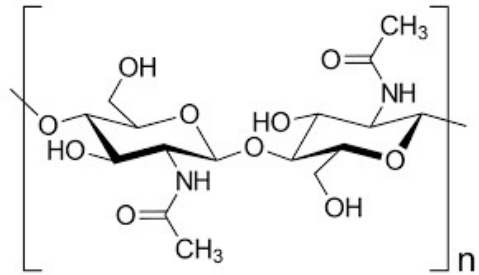
Naturally derived polymers are suitable for medical applications because of

- **Biocompatibility,**
- **Biodegradability,**
- **Non-toxicity,**
- **Ability to adsorb bioactive molecules.**

CLASSIFICATION OF NATURAL POLYMERS



EXAMPLE OF NATURAL POLYMER PRODUCTION/EXTRACTION



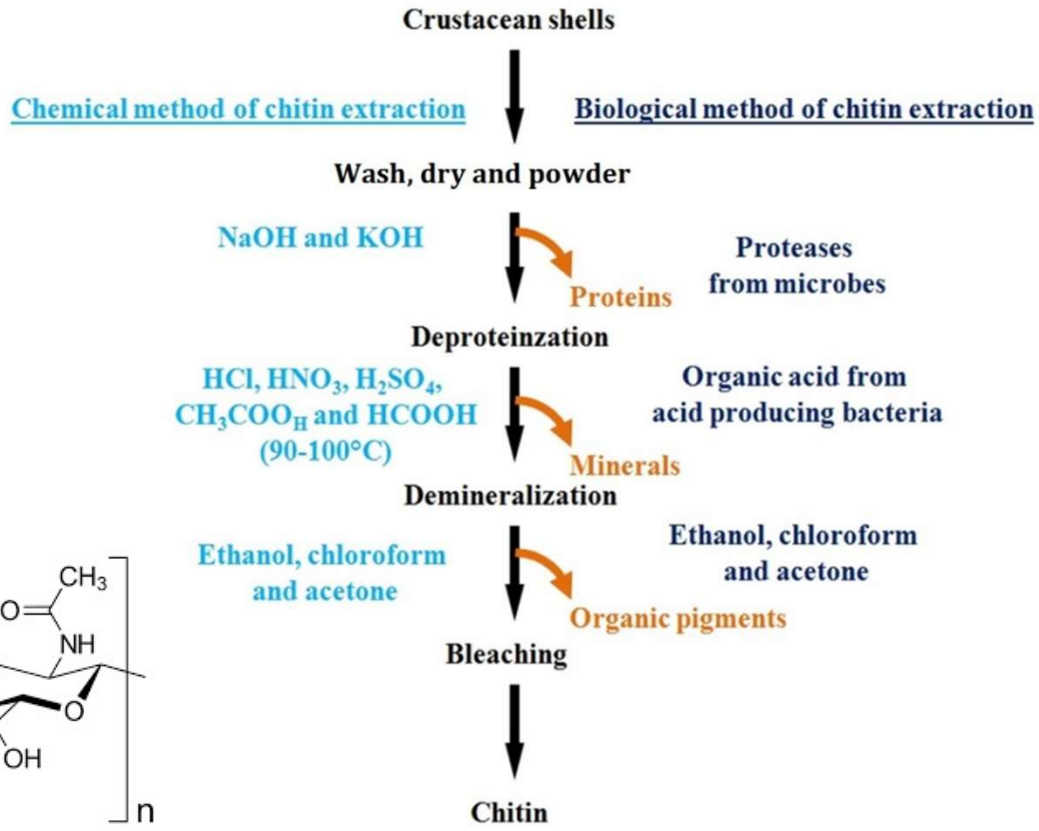
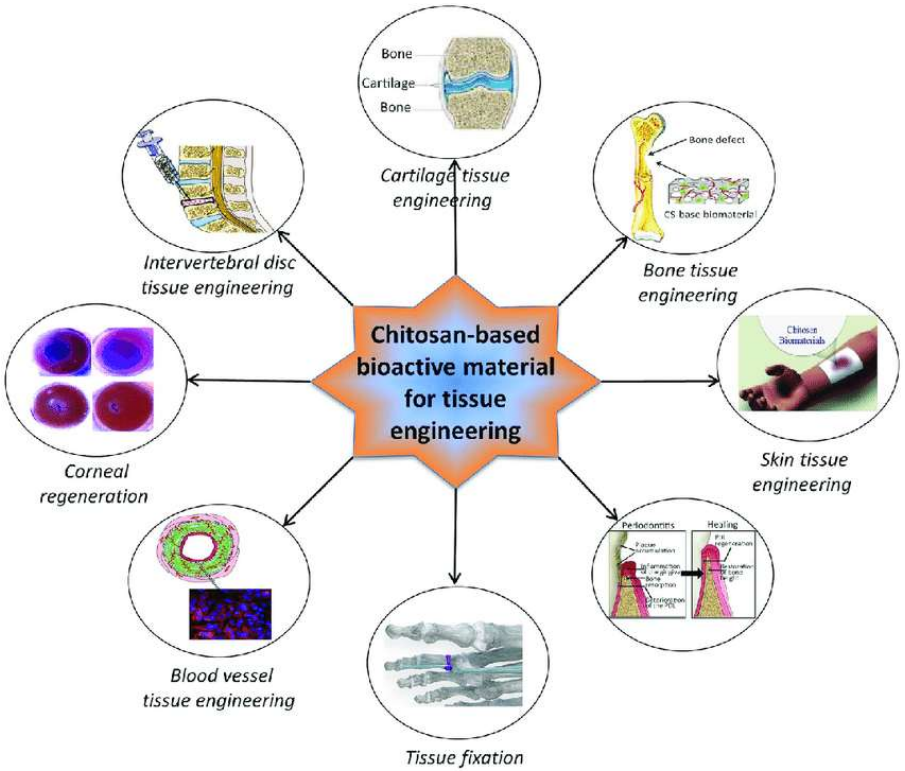




Figure:A schematic representation of the chemical and biological (enzymatic) methods for chitin extraction

P. Rameshthangam et al.,2018

CHITOSAN



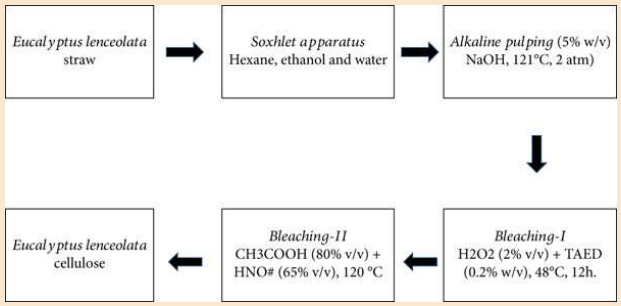
Derived from chitin, used in wound healing, drug delivery, and bone regeneration owing to its biocompatibility and antimicrobial properties. Its positive charge allows interactions with negatively charged biomolecules, enhancing bioactivity.

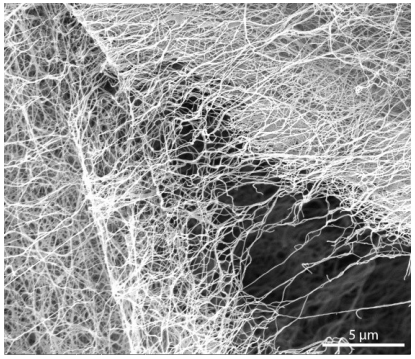
Advantages	Disadvantages
Biodegradable – Can be broken down by enzymes	Poor mechanical strength – Especially when hydrated
Antimicrobial activity – Can inhibit certain bacteria and fungi	Limited solubility – Poorly soluble at neutral pH
Hemostatic properties – Can help control bleeding	Variable antimicrobial activity – Depends on pH, molecular weight, and formulation
Good film and hydrogel formation – Suitable for wound dressings and scaffolds	Variable properties – Depend on source and degree of deacetylation

CELLULOSE

Properties

- Major sources of cellulose are plant fibers, bacteria.
- It is the most abundant natural polymer on Earth and is mainly obtained from plant cell walls (wood, cotton) and certain bacteria.
- Linear chains of glucose units linked by β-1,4-glycosidic bonds.





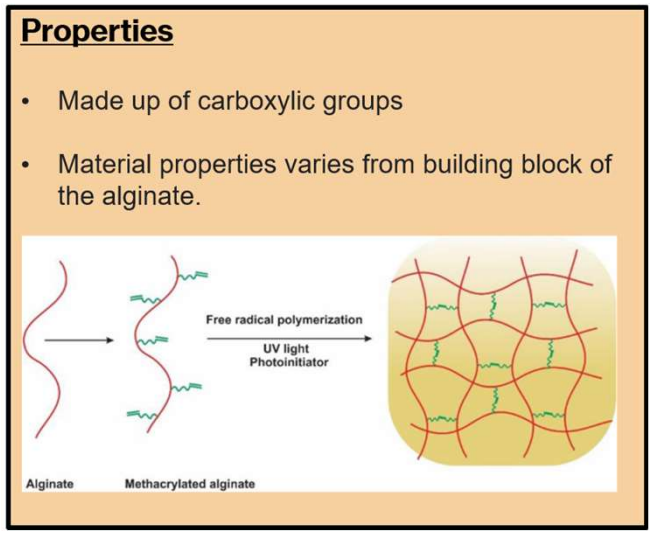
Bacterial Cellulose

Advantages	Disadvantages
Fibrous structure	Limited degradation in the body – Humans lack cellulase enzymes
Hydrophilic – Absorbs and retains moisture	Poor solubility – Difficult to dissolve in common solvents
Easy chemical modification – Hydroxyl groups allow functionalization	Processing difficulties – Complex dissolution and fabrication methods

Application

Wound dressing
Drug delivery
Tissue Engineering
Blood purification

ALGINATE





Alginate

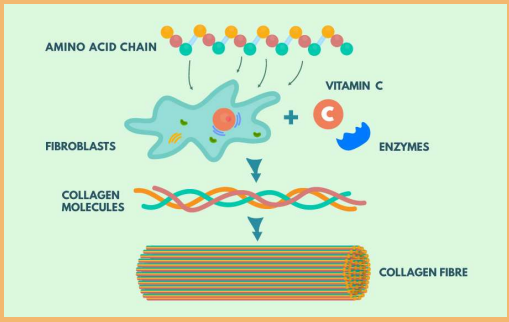
Advantages	Disadvantages
Biocompatible – Generally well tolerated by tissues	Poor mechanical strength – Hydrogels can be weak and brittle
Excellent hydrogel formation – Easily crosslinked using calcium ions	Poor cell adhesion – Lacks natural mammalian cell-binding sites
Hydrophilic – Absorbs and retains large amounts of water	Limited biodegradation in humans – No naturally occurring human alginate lyase
Mild processing conditions – Suitable for encapsulating living cells	Uncontrolled swelling – Can reduce structural stability
Renewable and abundant – Extracted mainly from brown algae	Variable properties – Depend on seaweed source, purity, and M/G ratio

- **Alginate**: Alginate is a naturally occurring, biocompatible, hydrophilic polysaccharide mainly extracted from brown seaweeds (Laminaria, Macrocystis, and Ascophyllum) and produced by certain bacteria. Alginate is applied in cartilage regeneration and wound care products. It forms hydrogels that are ideal for encapsulating cells in tissue engineering (Lee & Mooney, 2012).

COLLAGEN

Properties

- Collagen is a natural, biodegradable, biocompatible fibrous protein and the major structural component of the extracellular matrix (ECM) in connective tissues such as skin, bone, cartilage, tendons, and ligaments.
- Made up of amino acids group: Glycine, Proline, Hydroxyproline





Collagen

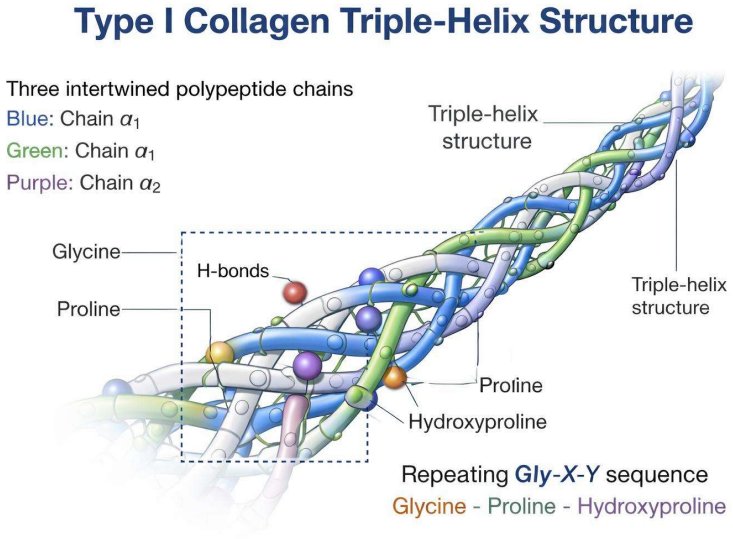
Advantages	Disadvantages
Biodegradable – Broken down by enzymes	Poor mechanical strength – Particularly in hydrated scaffolds
High bioactivity – Promotes cell adhesion and tissue regeneration	Rapid degradation – May require crosslinking
Hydrophilic – Absorbs and retains water	Low thermal stability – Sensitive to elevated temperatures
Natural ECM component – Mimics the biological environment of cells	Immunogenicity and variability – Depend on source and purification

Application

Scaffold as tissue filler
Support matrix for matrix –rich tissues

COLLAGEN

- Consist of three intertwined protein chains, helical structure
- Collagen…..non-toxic, minimal immune response
- Can be processed into a variety formats
  - Porous sponges, Gels, and Sheets
- Applications
  - Surgery, Drug delivery, Prosthetic implants and tissue-engineering of multiple organs

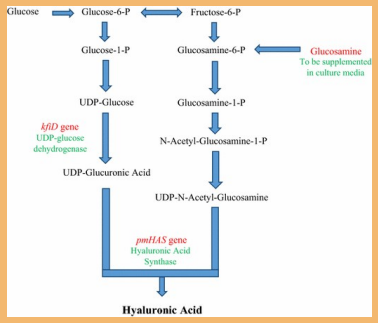


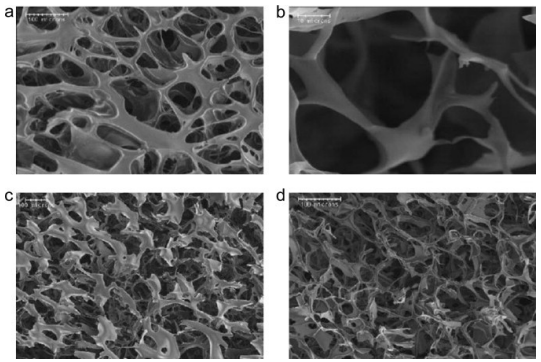
HYALURONIC ACID

Properties

Hyaluronic acid (HA), also known as hyaluronan, is a naturally occurring, biodegradable, biocompatible, and highly hydrophilic polysaccharide belonging to the glycosaminoglycan (GAG) family.
It is an important component of the extracellular matrix (ECM) and is naturally present in skin, cartilage, synovial fluid, and the vitreous humor of the eye.

β-D glucoronic acid and
β-1,3-N-acetyl-D-glucosamide.





Hyaluronic Acid

Application

Lubricant in the joints and other tissues
Skin treatment
Hydration, Hydrogel

Advantages	Disadvantages
High water retention – Maintains tissue hydration and lubrication	Poor mechanical strength – Uncrosslinked hydrogels are mechanically weak
Bioactive – Interacts with cell receptors and supports tissue repair	Rapid degradation – May require crosslinking for longer-term use
Viscoelastic – Provides lubrication and shock absorption	High production cost – Purification and medical-grade processing can be expensive

ADVANTAGES & DISADVANTAGES

Major advantages of natural polymers

Lower/no toxicity,

Better bioactivity,

Enhanced cell response when associated with cells,

Excellent biocompatibility,

Extreme hydrophilicity and effective biological function.

Significant drawbacks of natural polymers

• Complicated isolation techniques from inconsistent sources.

• Poor processability and solubility blocking the utilization of industrial fabrication processes.

• Possibility of contamination by pyrogens and pathogens.

Poor or limited material properties like elasticity, ductility, strength, and shelf life.

Immunogenic & chance to allergic reaction.

Low mechanical properties and easily degradable.

• High cost.

SYNTHETIC POLYMERS

WHY DO WE PREFER SYNTHETIC ONES?

- Tailor-able properties
- Predictable lot-to-lot uniformity
- Free from concerns of immunogenicity
- Reliable source of raw materials

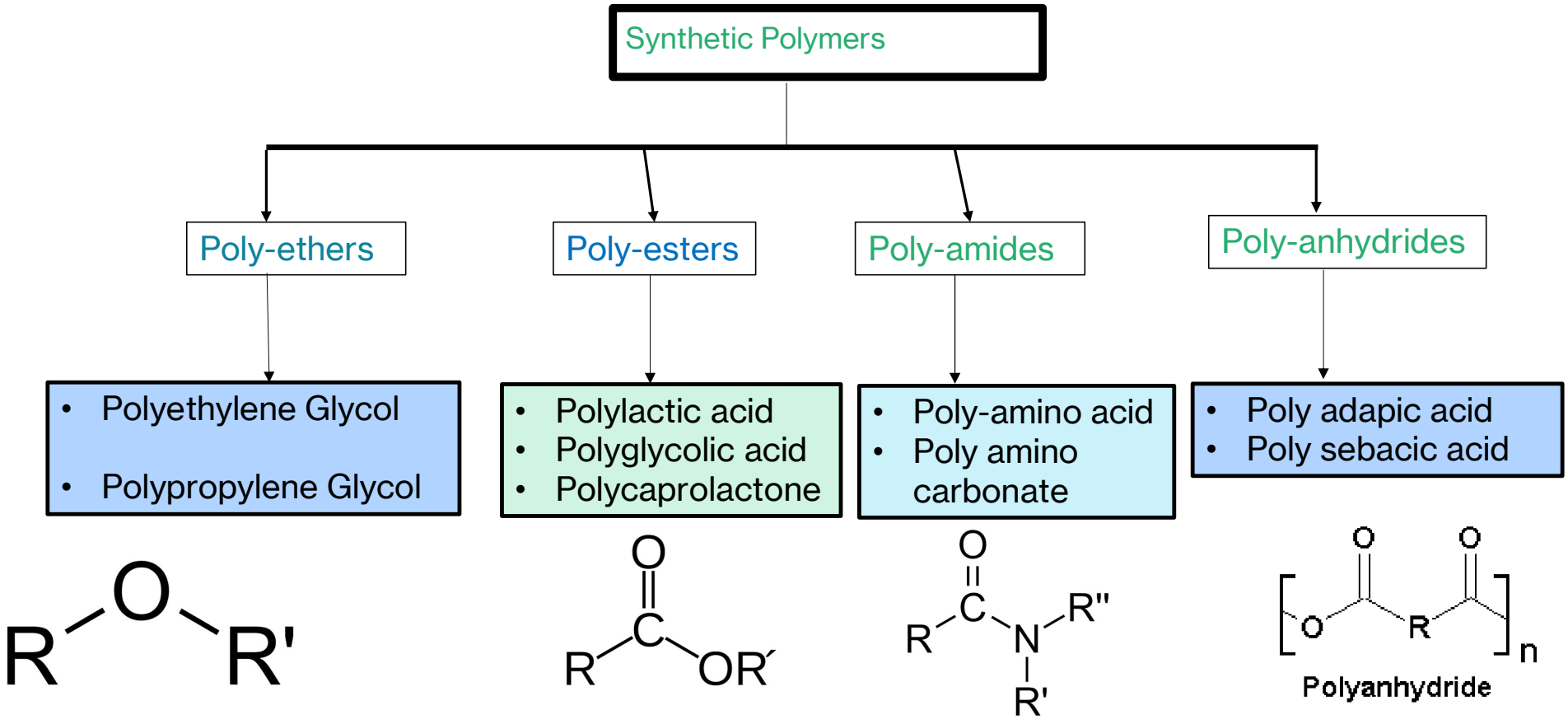
Synthetic polymers are man-made polymers produced by chemical reactions. Synthetic polymers have been used for numerous biomedical and pharmacologic purposes.

These include **prosthetic implants, suture material and drug carriers etc.**

Synthetic polymers :

- Polyvinyl chloride (PVC), polypropylene (PP),
- Polyethylene (PE),
- Polystyrene (PS), nylon,
- Polyethylene terephthalate (PET),
- Polyimide (PA),
- Polycarbonate (PC),
- Acrylonitrile butadiene (ABS), polyetheretherketone (PEEK)
- Polyurethane (PU).

CLASSIFICATION OF SYNTHETIC POLYMERS

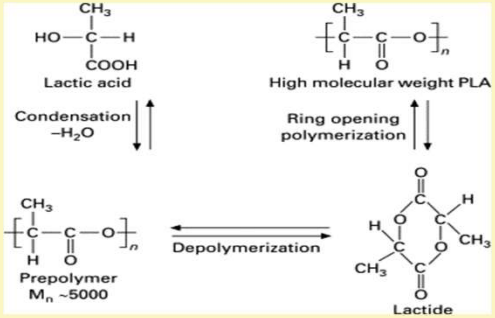


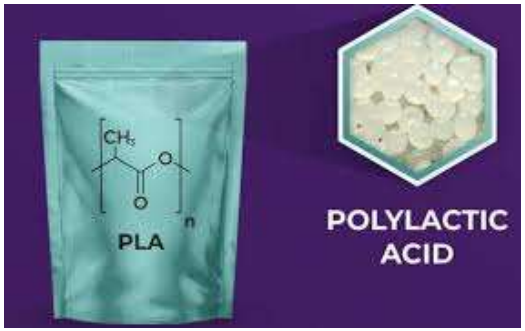
POLY-LACTIC ACID (PLA)

Properties

Linear highly crystaline polyester

PLA is produced from renewable resources such as corn starch and sugarcane and degrades mainly through hydrolysis of its ester bonds.





Poly Lactic Acid

Advantages	Disadvantages
Biocompatible – Generally well tolerated by biological tissues	Brittle – Low flexibility and toughness
Biodegradable – Breaks down into lactic acid	Slow degradation – May take months to years
Good mechanical strength – Suitable for selected temporary implants	Acidic degradation products – May cause local inflammation
Easy to process – Suitable for 3D printing and scaffold fabrication	Hydrophobic – Limits cell attachment

Application

Orthopedic fixation tools, ligament and tendon repair, vascular stents

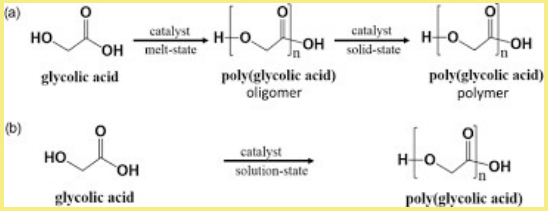
POLY-GLYCOLIC ACID (PGA)

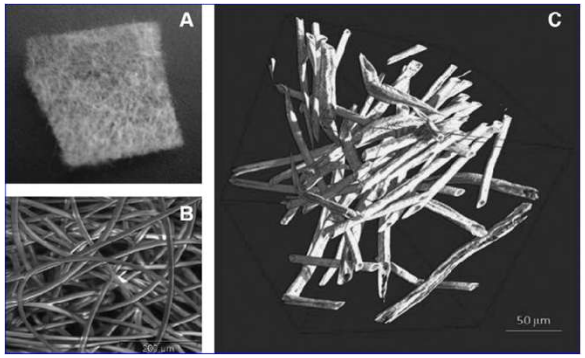
Properties

Highly crystaline

Ring-opening polymerization is preferred for biomedical-grade PGA because it can produce high-molecular-weight polymers with better mechanical properties.

A synthetic homopolymer of glycolic (hydroacetic) acid





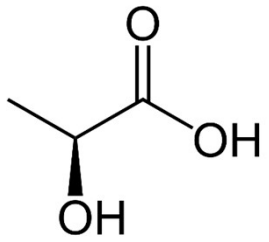
Poly-glycolic Acid

Advantages	Disadvantages
Biocompatible – Generally well tolerated by biological tissues	Brittle – Low flexibility and toughness
Biodegradable – Breaks down into glycolic acid	Rapid degradation – May lose mechanical strength before tissue healing is complete
High mechanical strength – Good initial tensile strength and stiffness	Acidic degradation products – May cause local inflammation
Relatively hydrophilic – Absorbs water more readily than PLA	Moisture-sensitive – Hydrolysis can reduce mechanical properties during storage and use
Bioabsorbable – Suitable for temporary medical devices	Difficult processing – High melting temperature and sensitivity to thermal degradation

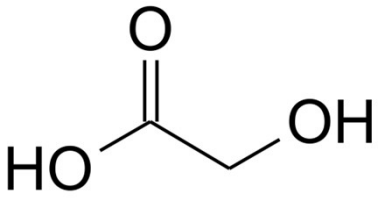
PLA, PLGA, AND PGA

PLA: **Hydrophobic**, meaning it repels water, Biodegrades through hydrolysis

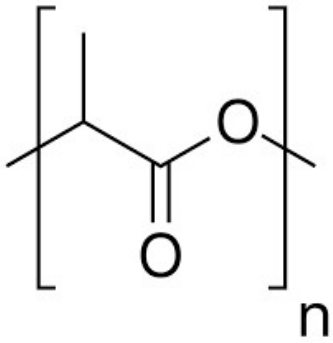
PGA: **Hydrophilic**, meaning it absorbs water, which accelerates its degradation, PGA degrades much faster than PLA.

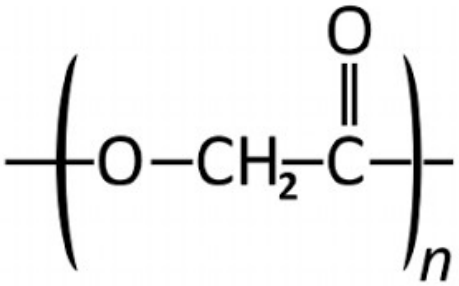


Lactic acid



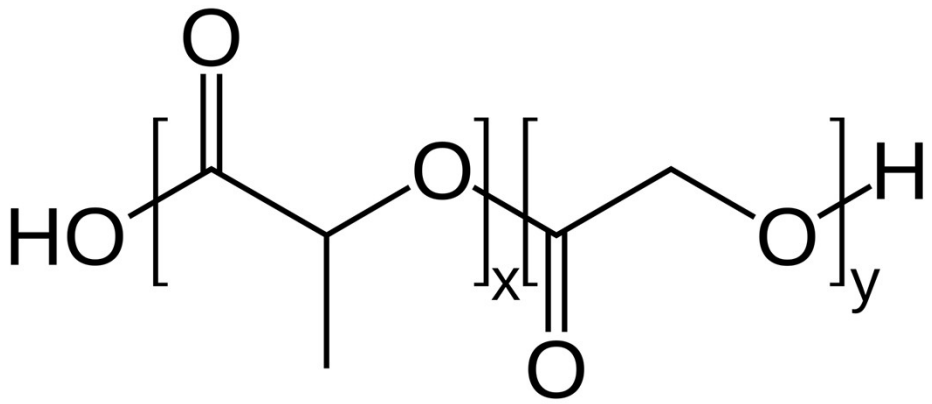
Glycolic acid





PLA, PLGA, AND PGA

Poly(lactic-*co*-glycolic) (PLGA) acid is a copolymer which is used in many medical devices, such as, grafts, sutures, implants, prosthetic devices, surgical sealant films, micro and nanoparticles.



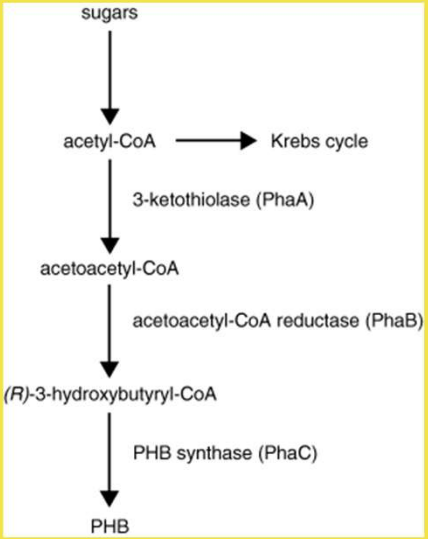
Depending on the ratio of lactide to glycolide used for the polymerization, different forms of PLGA can be obtained.

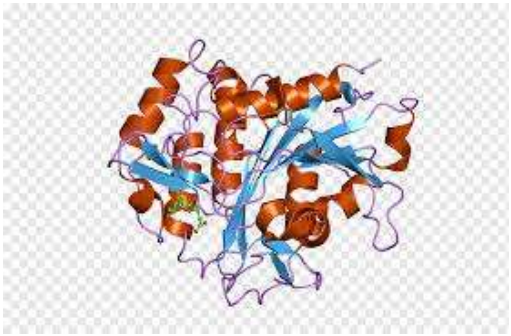
PLGA degrades by hydrolysis of its ester linkages in the presence of water.

PLGA has adjustable biodegradability

POLY-HYDROXYBUTYRATE ACID (PHB)

Properties
Naturally occuring b-hydroxy acid;
It is a homopolymer having a stereoregular structure with high crystalinity.





Poly Hydroxy-Butyrate

Advantages	Disadvantages
Biocompatible – Suitable for selected biomedical applications	Brittle – Low flexibility and impact resistance
Biodegradable – Can break down into 3-hydroxybutyrate	Slow degradation – High crystallinity can delay degradation
Naturally produced – Synthesized by microorganisms	High production cost – Fermentation and purification are relatively expensive
Good mechanical stiffness – Useful for structural scaffolds	Poor thermal stability – May degrade during melt processing
Biologically derived degradation product – 3-hydroxybutyrate is a naturally occurring metabolite	Limited processability – Narrow thermal processing window

POLYVINYL ALCOHOL (PVA)

Properties

- Semicrystalline polyhydroxypolymer Polyvinyl alcohol (PVA) is a synthetic, water-soluble, hydrophilic, and generally biocompatible polymer widely used in biomedical applications such as wound dressings, drug delivery systems, hydrogels, tissue engineering scaffolds, and contact lenses. Unlike PLA, PGA, PLGA, and PHB, PVA has a carbon–carbon polymer backbone and is not readily biodegradable under typical physiological conditions.

- Prepared via hydrolysis of poly vinyl acetate



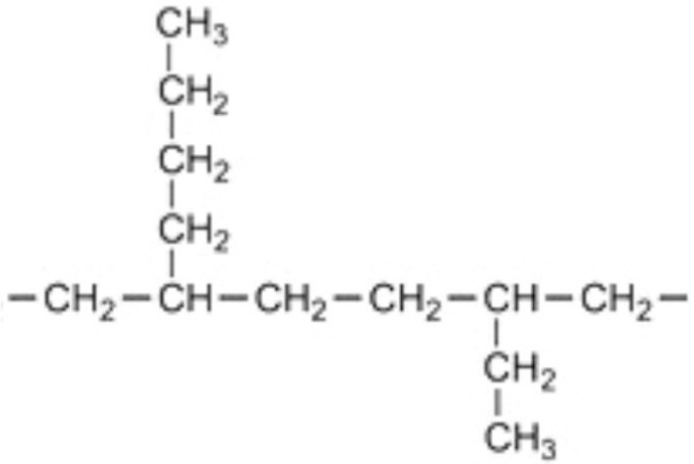


Polyvinyl alcohol

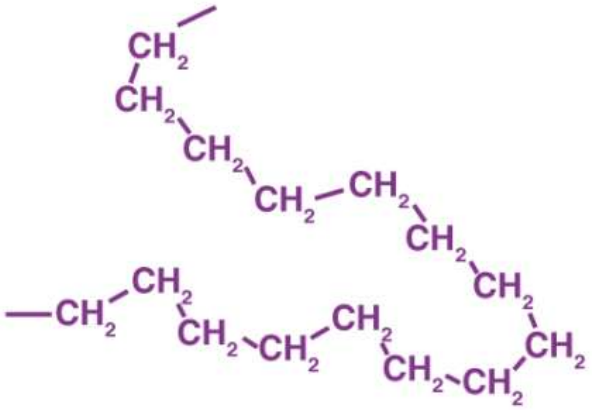
Advantages	Disadvantages
Biocompatible – Generally well tolerated by tissues	Poor biodegradability – Persists under many physiological conditions
Hydrophilic – Absorbs and retains water	Low wet mechanical strength – May soften or swell excessively
Hydrogel formation – Suitable for wound dressings and drug delivery	Water sensitivity – Uncrosslinked PVA may dissolve in aqueous environments
Low cost and easy processing – Can be fabricated into fibers, films, and gels	Limited cell adhesion – Often requires biological modification

POLYETHYLENE (PE)

Low-Density Polyethylene (LDPE)



High-Density Polyethylene (HDPE)

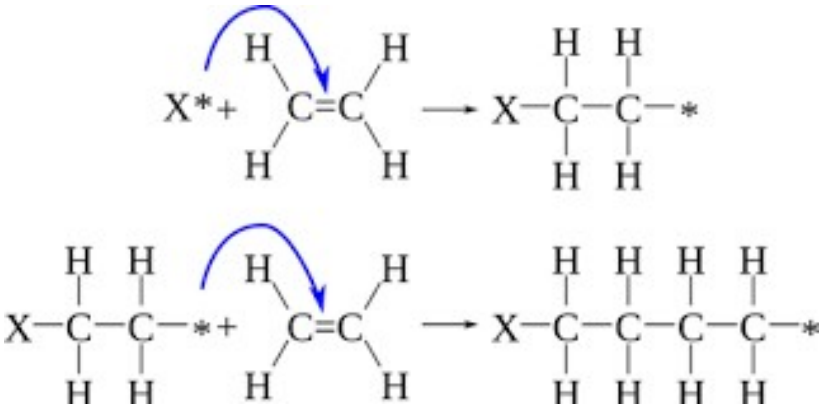


Low-Density Polyethylene (LDPE): LDPE has relatively short and branched polymer chains. It is more amorphous and has a lower density, which creates a flexible and soft material.
High-Density Polyethylene (HDPE): HDPE has longer and more linear polymer chains.

For high mechanical strength, PE has to be linear.

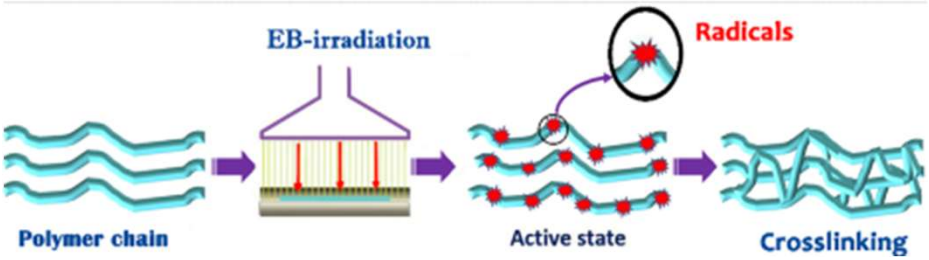
ULTRA-HIGH MOLECULAR WEIGHT POLYETHYLENE

UHMWPE (Ultra-High Molecular Weight Polyethylene) has an **extremely high molecular weight**, typically in the range of **3 to 6 million g/mol**, which gives it different properties.



free radical initiator (X•)

Gamma irradiation can induce **cross-linking** between polymer chains. This means that individual polymer chains are chemically bonded together at certain points, forming a more interconnected and robust network.



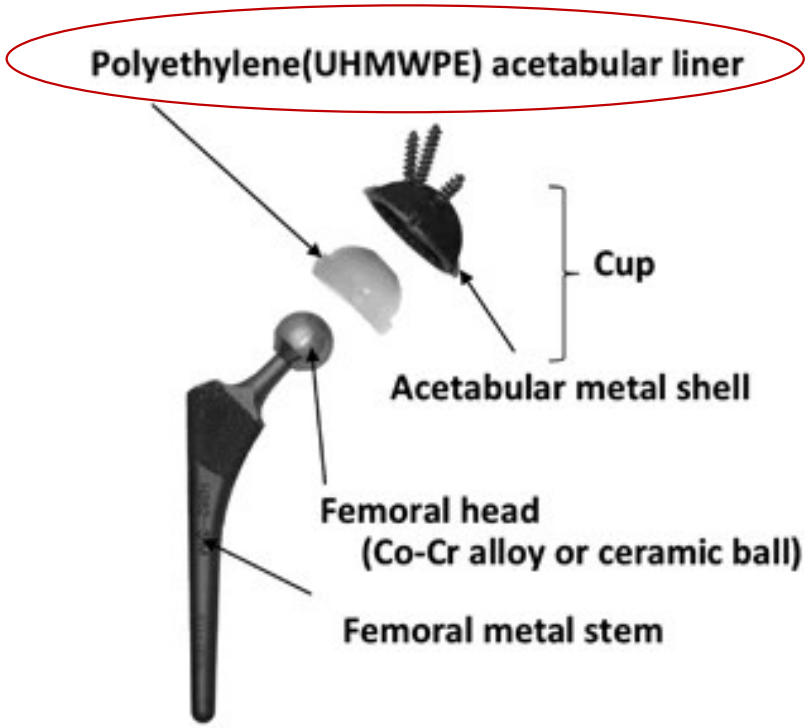
The polymer chains in UHMWPE are significantly longer than those in HDPE, which gives the material extraordinary strength and toughness.

ULTRA-HIGH MOLECULAR WEIGHT POLYETHYLENE

Due to its very high molecular weight, UHMWPE has **exceptional abrasion resistance**, **impact strength**, and **low friction**.

Ultra-high molecular weight polyethylene (UHMWPE) offer excellent mechanical properties similar to natural bone, high impact, wear resistance, low friction coefficient, and biocompatibility for prostheses, they lack the ability to directly bond with bone tissue. For this reason, UHMWPE is widely used to develop knee, hip, and shoulder prostheses.

Attempts to make UHMWPE bioactive have been made by adding hydroxyapatite (HA) to it.



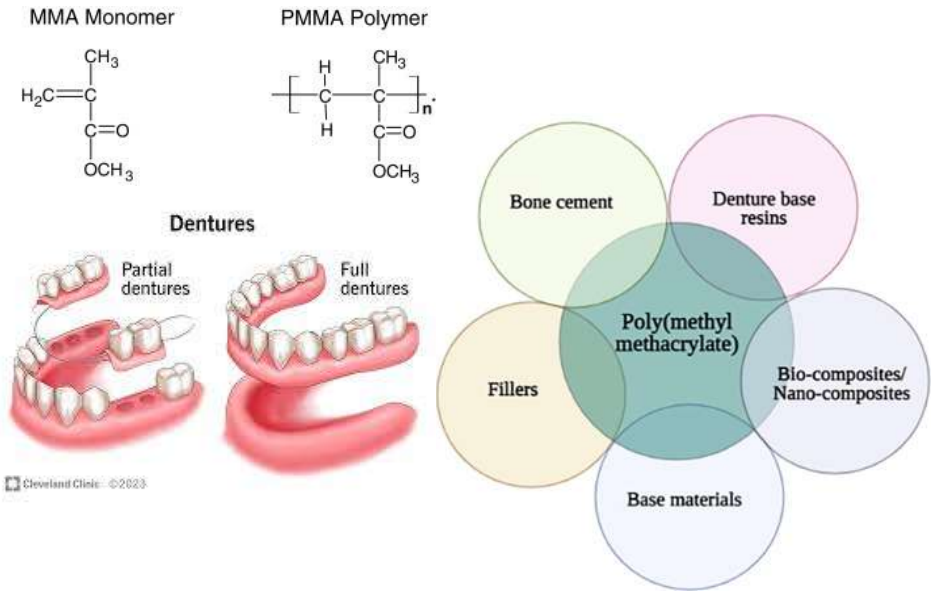
Artificial hip joint system

https://doi.org/10.1038/pj.2015.45

POLYMETHYLMETACRYLATE (PMMA)

Polymethyl methacrylate (PMMA) is a synthetic, transparent, biocompatible, non-biodegradable thermoplastic polymer widely used in biomedical applications.

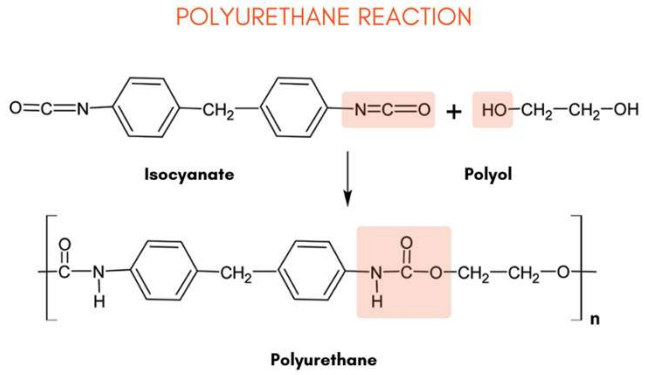
Prosthesis applications require fixation to bone that relies on using poly (methyl methacrylate) (PMMA) as a bone cement, which provides initial fixation but hinders osseointegration, the crucial process of bone growth onto the implant



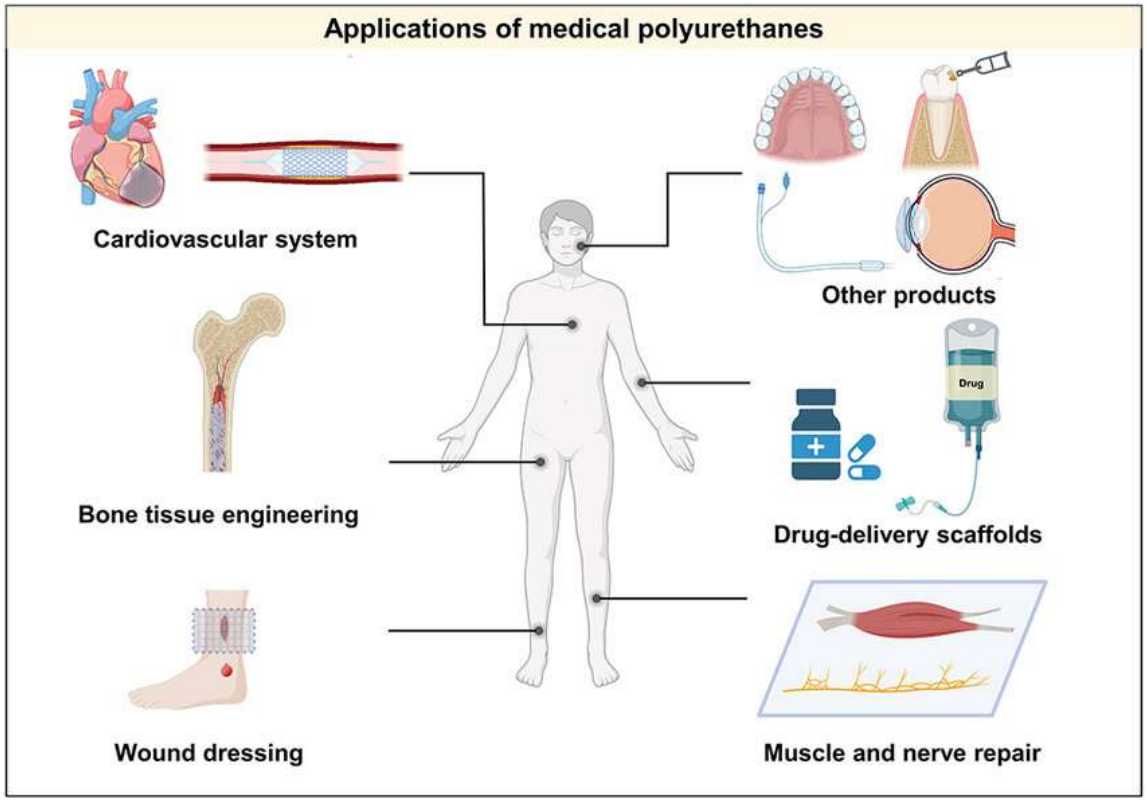
Advantages	Disadvantages
Biocompatible – Well tolerated after proper polymerization	Non-biodegradable – Remains permanently in the body
Good mechanical strength – High stiffness and compressive resistance	Brittle – Low impact resistance and fracture toughness
Optically transparent – Suitable for ophthalmic applications	Residual monomer toxicity – Unreacted MMA may irritate tissues
Easy to process – Can be molded, cast, and polymerized in situ	Heat generation – Polymerization is exothermic and may damage surrounding tissues
Good dimensional stability – Maintains shape over time	Poor bioactivity – Does not naturally bond directly with bone

POLYURETHANE

Polyurethane (PU) is mainly composed of diisocyanate as hard chain segments and oligomeric polyols as soft chain segments.



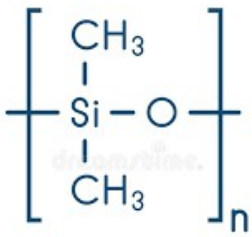
The physical properties of **polyurethane** (PU) can vary significantly based on the **synthesis conditions**, particularly the **degree of cross-linking**.
PU is biocompatible and have low water permeability, but disadvantages is isocyanonides have toxicity issues.



SİLİCONES

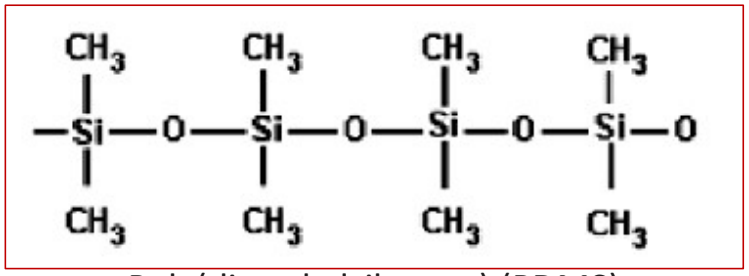
Polymer backbone is composed of alternating silicon and oxygen atoms.

Siloxane is the basic repating unit



polydimethylsiloxane

Silicones are water repellent, heat stable, and very resistant to chemical attack



Poly(dimethylsiloxane) (PDMS)

Advantages

- Low T_g
- Environmental resistance
- Bioinert
- Highly flexible
- Good blood compatibility
- Electrically insulating
- Non-flammable

Disadvantages

- Low mechanical modulus
- Hydrolysis susceptible
- Attack by acids and bases
- Poor abrasion
- High protein absorption

In both blood contacting and ophthalmic applications, the ability to prevent the nonspecific adsorption of proteins is essential for proper function. So silicones have a problem for that type of applications.

SİLİCONES

- Silicone elastomer
  - Widely used because it is strong, very mobile bone of their Si-O-Si (siloxane) caternary backbone; which provide chemical inertness and flexibility, stable over time at a body temp., show little tissue reactivity, and highly resistant to chemical attack and heat.

The use of silicone materials in medical implantable devices can be seen for example in facial, breast, pacemaker and cochlear implants.[6] However, the applications of silicone elastomer are often limited due to the inherently **poor mechanical properties** of these materials, particularly in relation to tensile and tear strength, abrasion, and flex-fatigue life.

SİLİCONES

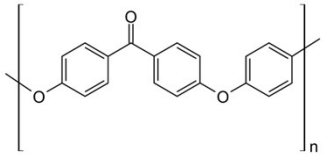
Another disadvantage of conventional silicone elastomers in device manufacturing is the need for crosslinking to develop useful properties. This crosslinking process is needed to form chemical bonds among adjacent polymer chains in order to gain rubber elasticity and the physical-mechanical properties required. However, once crosslinked, the resulting thermoset silicone cannot be redissolved or remelted. This reduces the number of post-fabrication operations that can be used in device manufacturing with these silicones, relative to those possible with thermoplastic biomaterials such as thermal forming, tipping, and tapering, radio-frequency welding, heat sealing and solvent bonding.

POLY (ETHER-ETHER-KETONE)

Polyether ether ketone (PEEK) is a high-performance, semi-crystalline thermoplastic polymer belonging to the polyaryletherketone (PAEK) family.

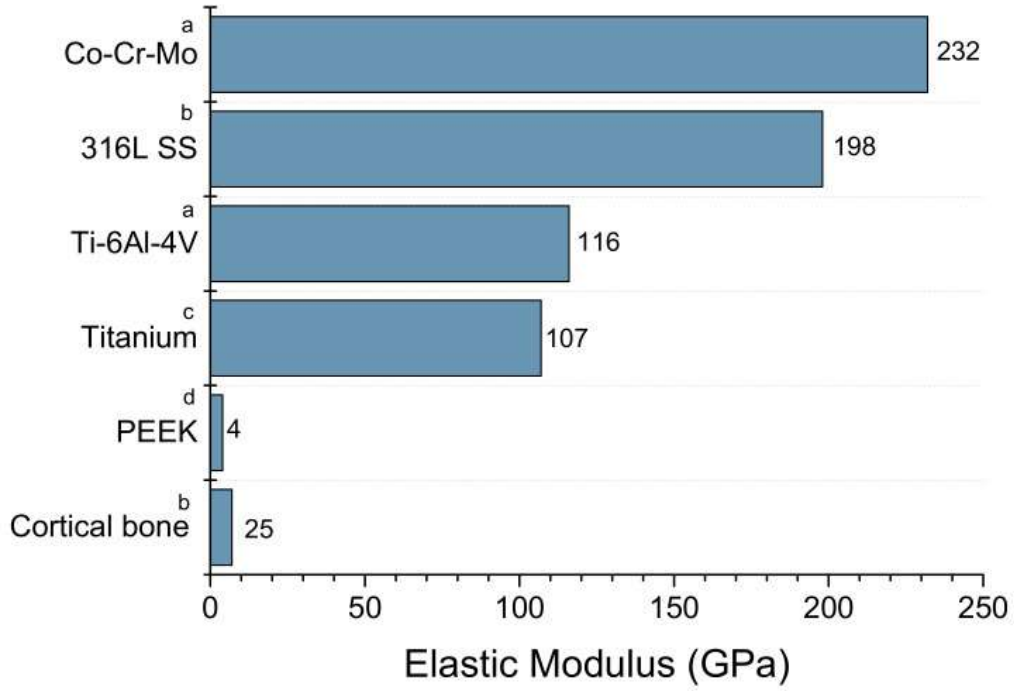
Aromatic Rings: PEEK contains aromatic rings in its polymer chain, which provide strength and stiffness. These rings are more stable and less prone to deformation than aliphatic structures found in other polymers.

Crystallinity: PEEK has a crystalline structure, meaning its polymer chains are arranged in an ordered pattern. This arrangement contributes to its strength and stiffness.



Advantages	Disadvantages
Biocompatible – Suitable for selected long-term implants	Bioinert – Poor natural bone bonding and osseointegration
Good mechanical strength – High stiffness and toughness	Expensive – High material and manufacturing costs
Bone-compatible modulus – Lower stiffness than most implant metals	Non-biodegradable – Remains permanently in the body
Radiolucent – Allows clearer X-ray and CT imaging	Hydrophobic surface – Limits cell adhesion and integration
Excellent chemical and thermal stability – Resists degradation and repeated sterilization under validated conditions	High processing temperature – Requires specialized equipment

POLYETHER ETHER KETONE (PEEK)

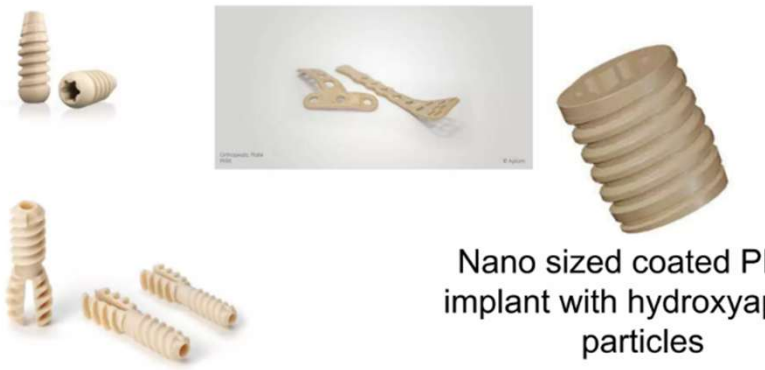


Elastic modulus of bone, PEEK, and metals usually used in implants

Dental implants
Dental implant abutments
Orthodontic wires
Prosthesis



Need improvement in properties



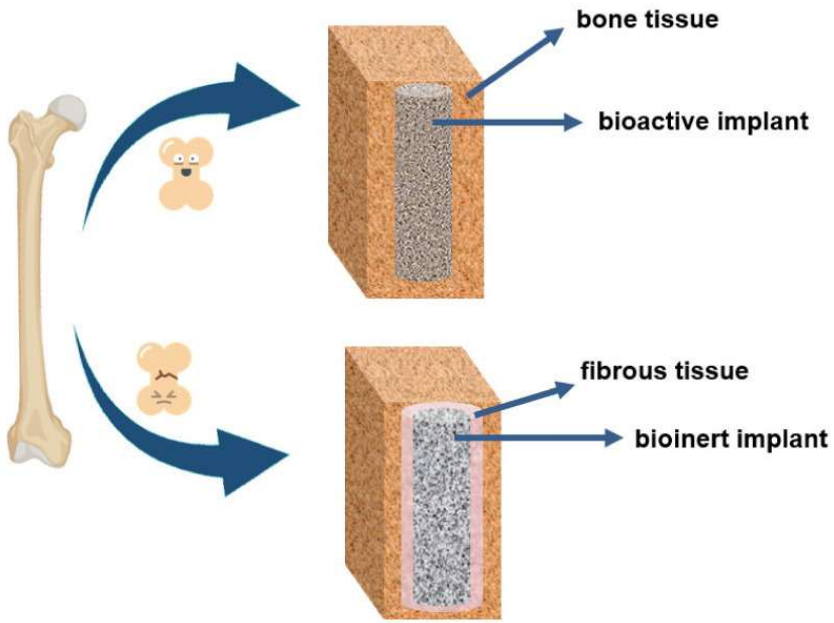
Nano sized coated PEEK implant with hydroxyapatite particles

PEEK

PEEK has hydrophobicity and lack of bioactivity.

Surface modification: surface modification technique aims to alter the surface characteristics of PEEK without affecting its bulk properties.

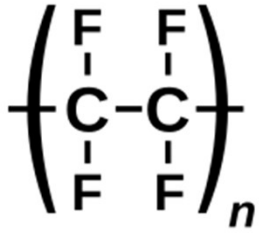
Bulk modification: compounding it with nanoparticles such as hydroxyapatite.



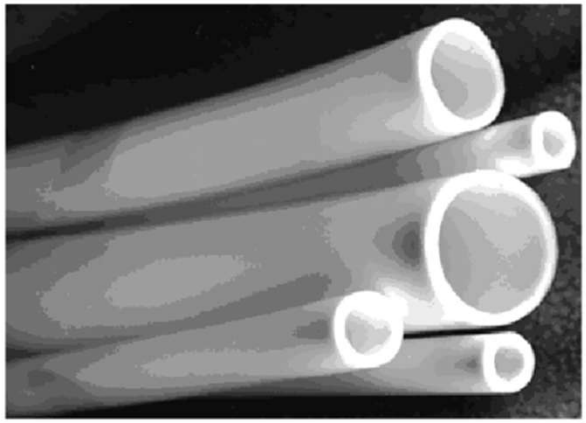
TEFLON

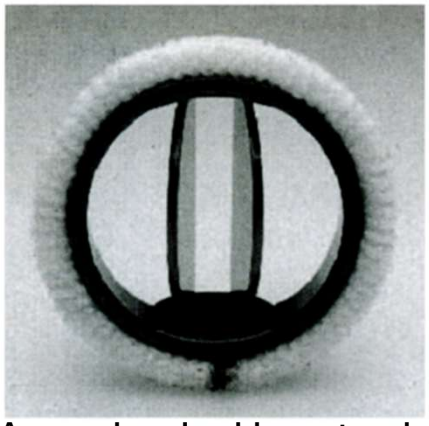
Polytetrafluoroethylene (PTFE), commonly known by the trade name Teflon, is a synthetic fluoropolymer characterized by excellent chemical resistance, low friction, hydrophobicity, and biostability.

PTFE is synthesized through free-radical addition polymerization of tetrafluoroethylene (TFE)



Advantages	Disadvantages
Chemically inert – Resistant to bodily fluids and chemicals	Non-biodegradable – Remains in the body permanently
Low friction – Excellent lubricating properties	Poor tissue integration – Low surface energy limits cell attachment
Hydrophobic – Very low water absorption	Poor surface adhesion – Difficult to bond or coat
	Difficult processing – Cannot be processed by conventional melt extrusion like many thermoplastics





A mechanical heart valve

PTFE is needed to ensure a medical device does not accumulate any bacteria during procedures but also has high non-stick properties as well.

Thermoplastic Polyurethane (TPU)

Thermoplastic polyurethane (TPU) is a synthetic, flexible, and elastomeric polymer widely used in biomedical applications due to its excellent elasticity

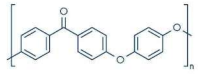
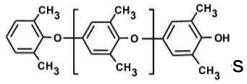
TPUs are very attractive candidates as biomaterials especially for the mimicking of soft tissue, due to their great flexibility in properties and ease of processing. Generally, TPUs have excellent physical properties, combining high elongation and high tensile strength to form tough, albeit fairly high-modulus elastomers.

Thermoplastic Polyurethanes (TPUs) generally have mechanical properties superior to silicone elastomers, are biocompatible and are now being commonly used as biomaterials.

This polymer shows good blood compatibility. It is also noncytotoxic and does not give rise to adverse tissue reactions.

Advantages	Disadvantages
Highly flexible – Excellent elasticity and elongation	Limited biodegradability – Most conventional TPUs remain stable in the body
Good mechanical strength – High toughness and tear resistance	Creep and stress relaxation – May deform during prolonged loading
Abrasion resistant – Suitable for moving and flexible devices	Processing sensitivity – Moisture and excessive heat can affect properties
Easy to process – Suitable for extrusion, injection molding, and 3D printing	Variable biocompatibility – Depends on additives, composition, and degradation products

SYNTHETIC POLYMERS

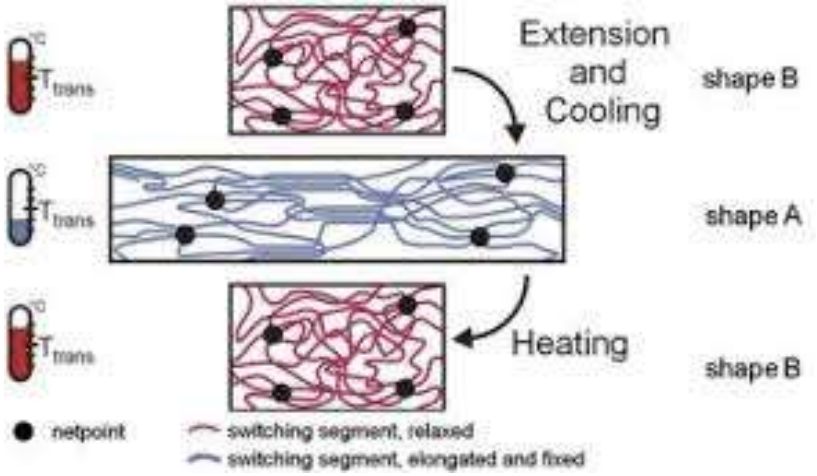
Polymer	Chemical Structure	Properties	Degradation	Common Additives	Examples	Structure-Function Relationship
Polylactic Acid (PLA)	[-C(CH3)HCOO-]n	Biodegradable, high strength	Hydrolysis to lactic acid	Plasticizers (e.g., PEG), antioxidants	Resorbable sutures, orthopedic screws	The methyl group (-CH3) increases hydrophobicity and crystallinity, improving mechanical strength and enabling slow, controlled hydrolysis.
Polyglycolic Acid (PGA)	[-CH2COO-]n	High crystallinity, fast degradation	Hydrolysis to glycolic acid	Copolymers with PLA (e.g., PLGA)	Tissue scaffolds, resorbable meshes	The simple, linear backbone allows tight packing, increasing crystallinity and the COO accelerates hydrolysis, ideal for fast resorption.
Polycaprolactone (PCL)	[-CH2(CH2)4COO-]n	Slow degradation, flexible	Hydrolytic degradation	Nanoparticles, bioactive ceramics	Long-term drug delivery systems	The long aliphatic chain provides flexibility and hydrophobicity, which slows degradation and makes it suitable for sustained release.
Polyether ether ketone (PEEK)		High strength, biostable	Non-degradable	Carbon fibers, TiO_2 nanoparticles	Spinal cages, dental implants	Aromatic rings offer rigidity and chemical resistance; ether and ketone groups balance flexibility and thermal stability.
Polyurethane (PU)		Elastic, durable	Varies (some degrade, others do not)	Antimicrobial agents, plasticizers	Vascular grafts, pacemaker leads	The segmented structure of hard and soft domains allows tuning of elasticity and strength, ideal for dynamic environments like blood vessels.

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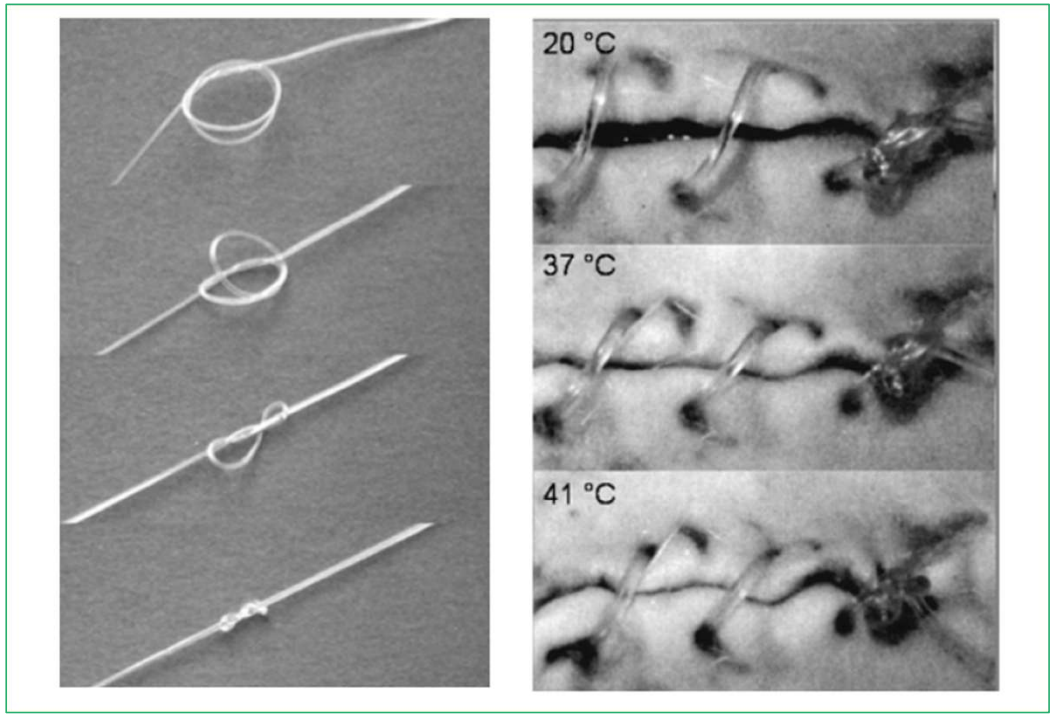
SHAPE MEMORY POLYMERS

The shape memory mechanism of polymers (SMPs) involves the ability of a polymer to "remember" its original shape and return to it when exposed to an external stimulus, such as heat.

thermal shape memory polymer



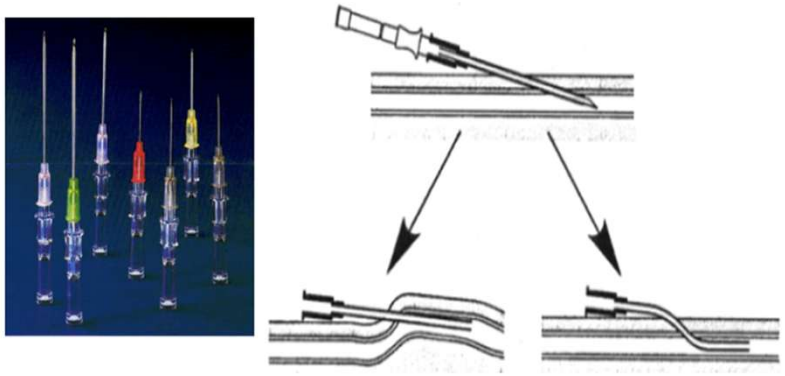
Shape memory cycle:
heating → deforming → cooling → reheating



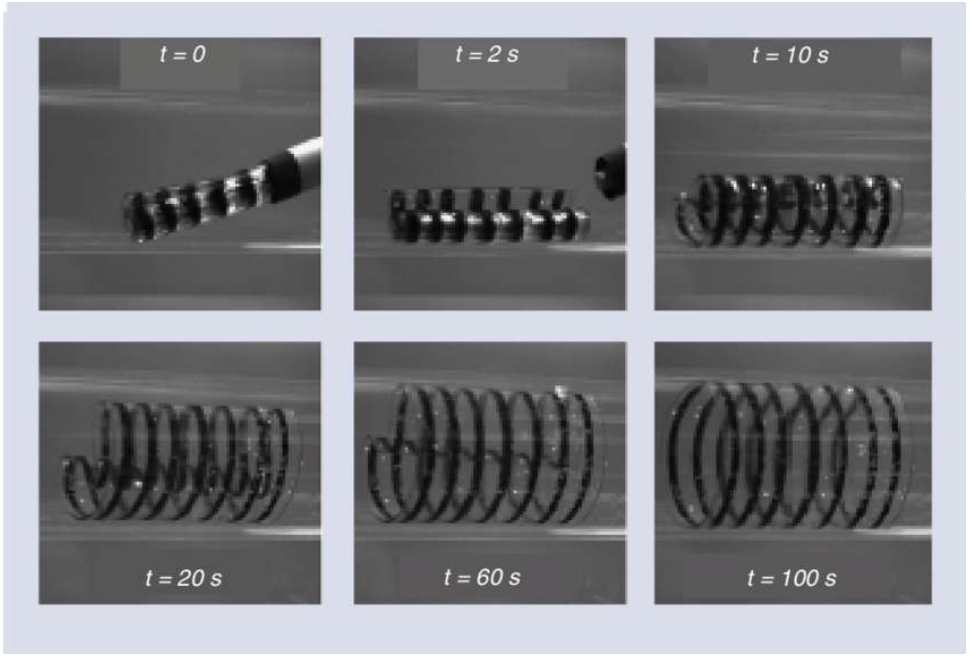
Suture becomes tightened with body temperature.

SHAPE MEMORY POLYMERS

Polymers that can recover deformation in response to a stimulus



Elastikleşen (yumuşayan) iğne



self-expandable stents

Polylactic acid is a SMP

DEGRADATION

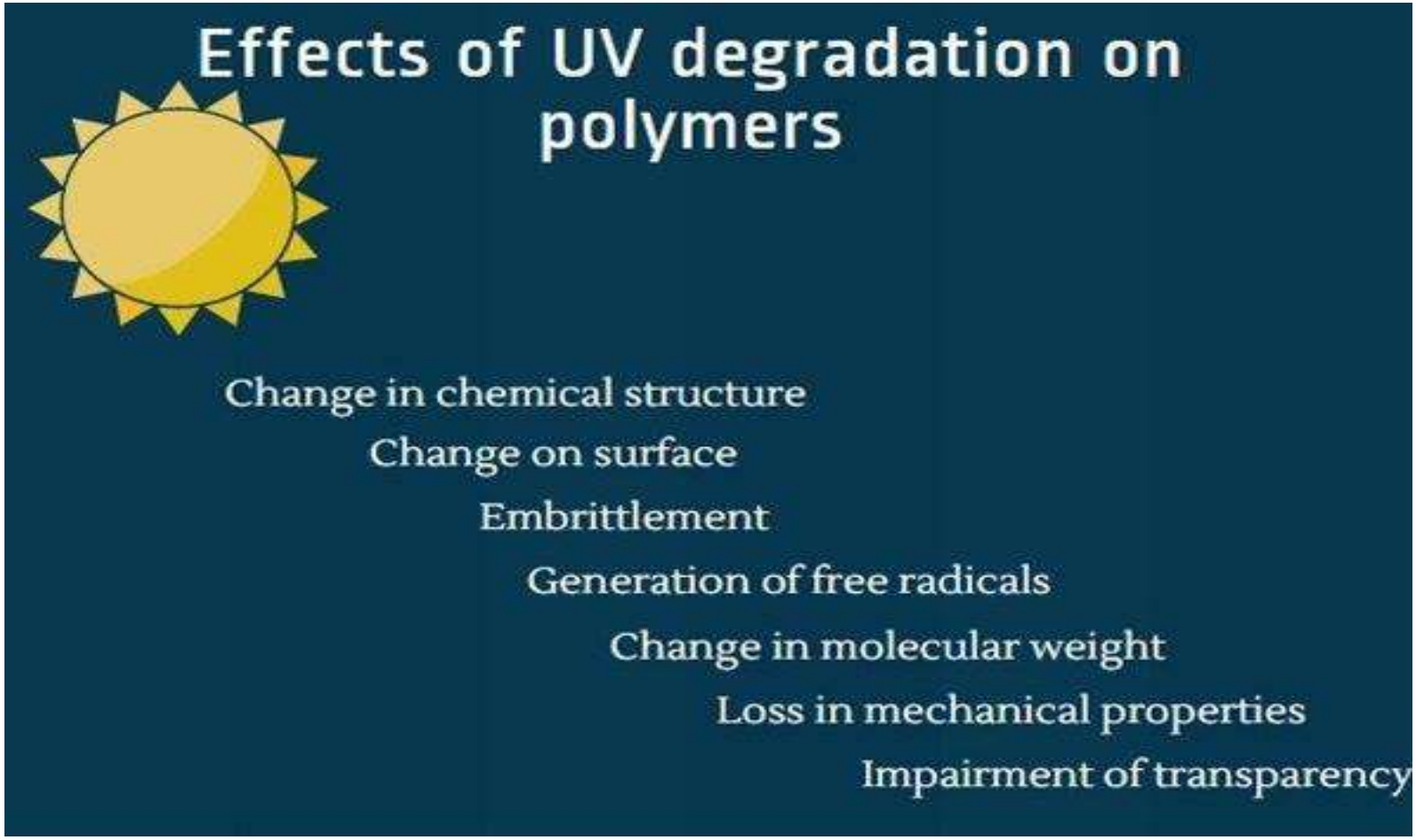
POLYMER DEGRADATION

Polymer degradation:- change of properties tensile strength, colour, shape and etc of polymer –based product under the influence of one or more environmental factors: heat, light or chemicals (acids/alkalis and salt)



Illustration of photo and thermal degradation of a shopping bag incorporating EPI's TDPA® Additive (top row) vs. a bag without EPI's TDPA™ Additive (bottom row). Test procedures follow ASTM D5272 'Outdoor Exposure Testing of Photo Degradable Plastics' Guidelines

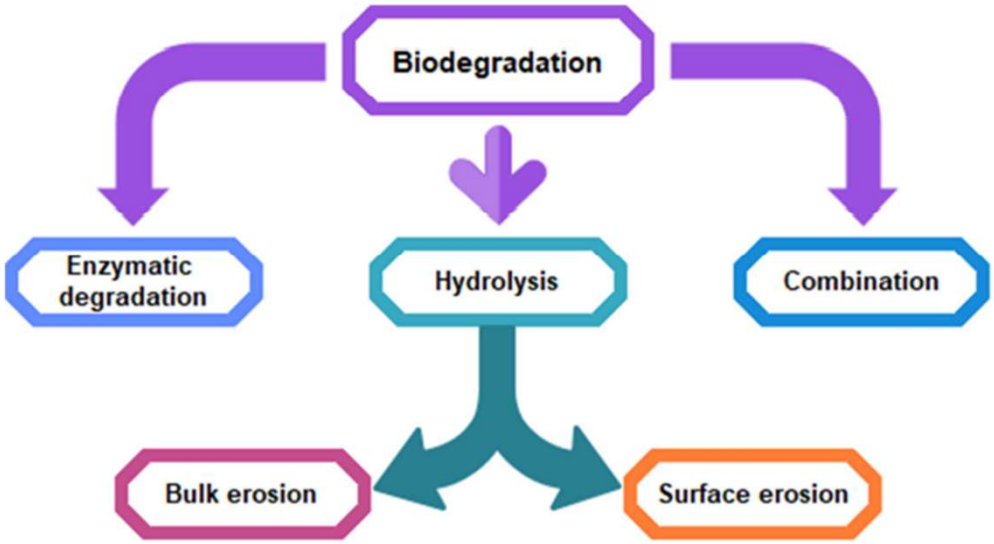
POLYMER DEGRADATION



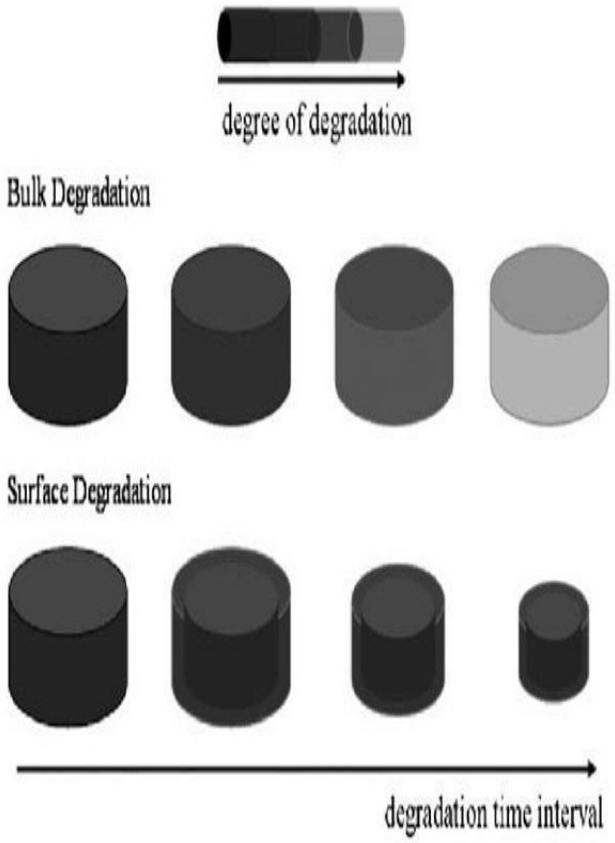
FACTORS AFFECTING DEGRADATION

- Morphological factors
  - Shape & size
  - Variation of diffusion coefficient and mechanical stresses
- Chemical factors
  - Chemical structure & composition
  - Presence of ionic group and configuration structure
  - Molecular weight and pressure of low molecular weight compounds
- Physical factors
  - Processing condition
  - Sterilization process

BIO-DEGRADABILITY OF POLYMERS



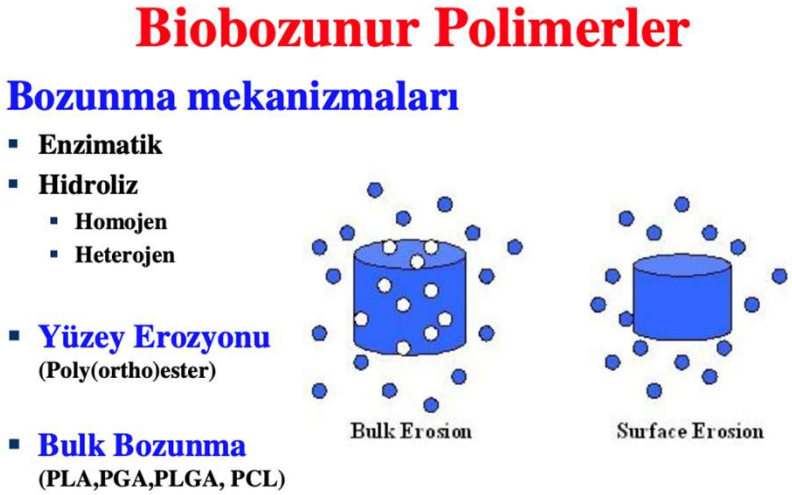
Biodegradability	
Enzymatic	**Hydrolytic**
Collagen, gelatin, silk fibroin, fibrinogen, elastin, laminin, soy protein, chitosan, alginate, hyaluronic acid, starch	Gelatin, cellulose acetate, PGA, PLLA, PLGA, PCL, PVA, PEO, PVP

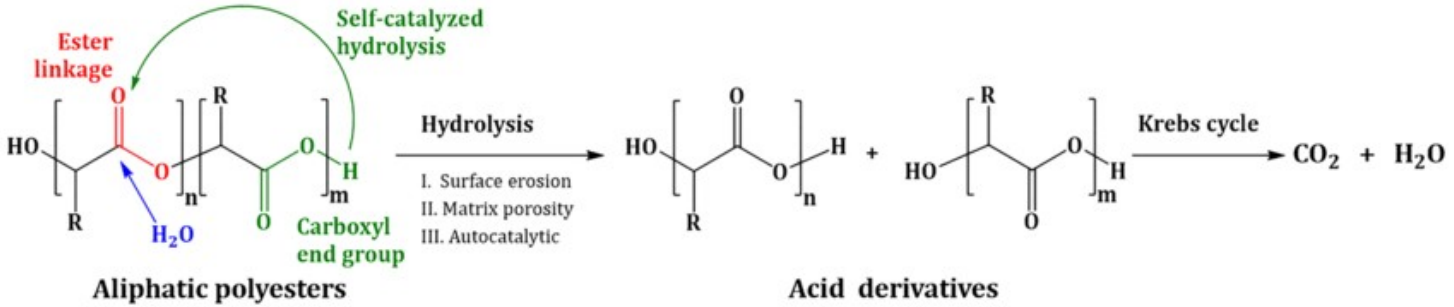


MSB Reddy et al., 2021

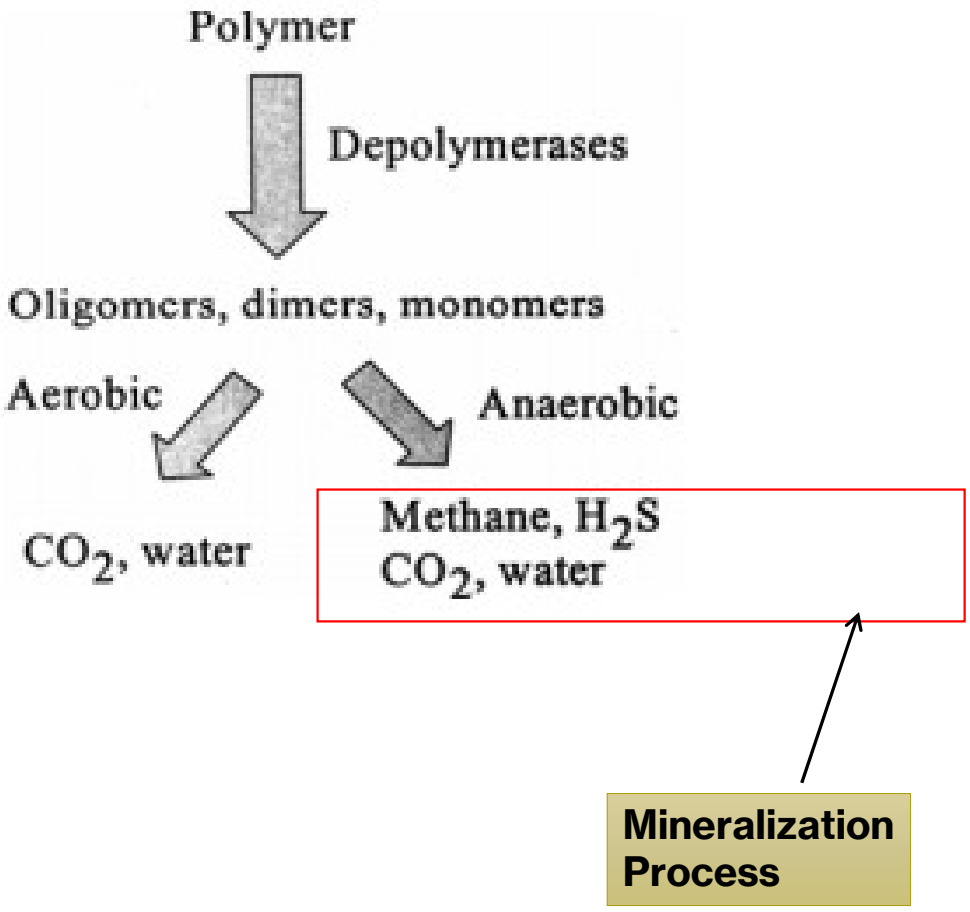
DEGRADATION OF POLYMERS

- **Hydrolytic Degradation**: Occurs primarily in aliphatic polyesters like PLA, PGA, and PLGA. Water breaks ester bonds, leading to predictable erosion.
- **Enzymatic Degradation**: Seen in natural polymers such as collagen and chitosan, broken down by proteolytic enzymes in vivo. This pathway supports cell-mediated remodeling and integration into host tissue.
- **Oxidative Degradation**: Common in polyurethanes and other polymers used in cardiovascular implants. Macrophage-mediated oxidative stress can cause surface cracking and embrittlement over time (Ratner et al., 2012).





REACTION PATHS OF POLYMER DEGRADATION

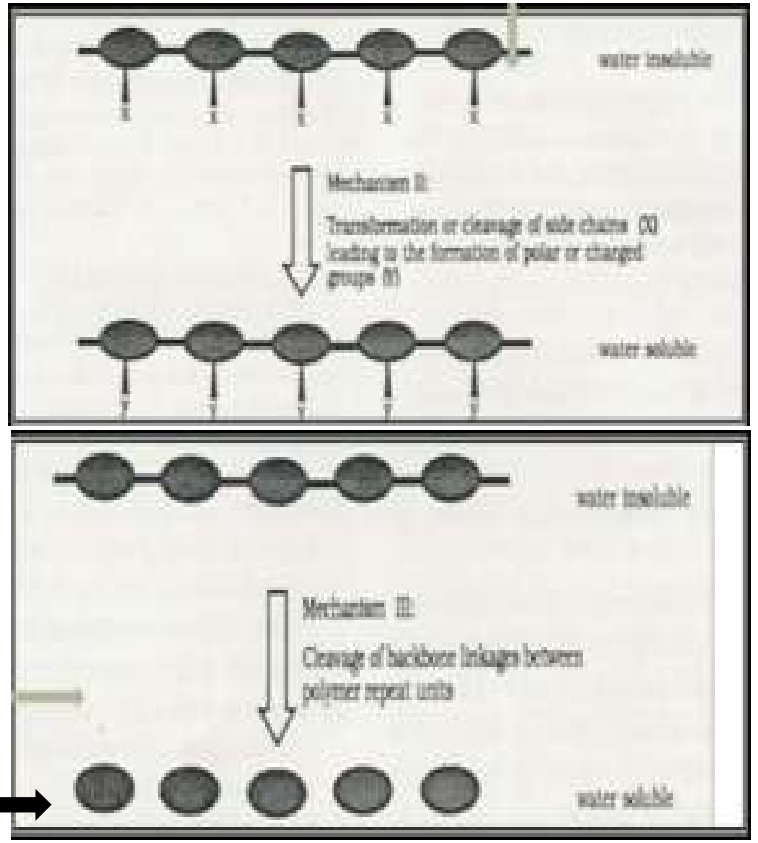


-**Small variations** of polymer **chemical structures** effects its **biodegradability**
-Biodegradability depend on **molecular weight, molecular form** and **crystallinity**
-**Increase** in **molecular weight** lead to **decrease** in **biodegradibility**
-**Enzymes** (**extracellular** & **Intrcellular** depolymerases) involved in **depolymerization** process

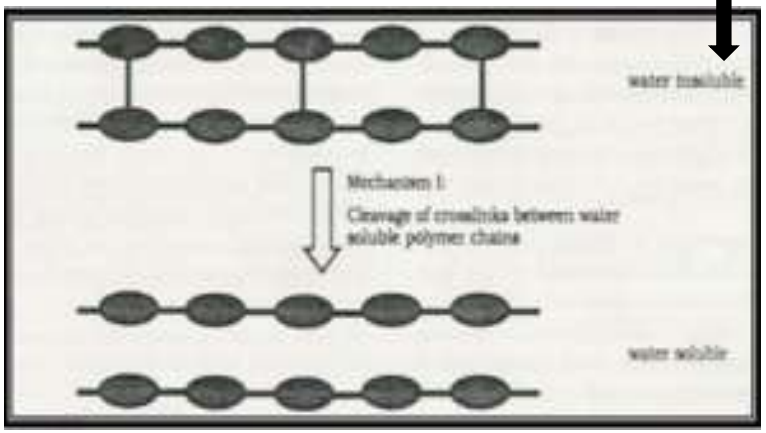
ENZYMATIC DEGRADATION

- **Enzymatic degradation –** mediated by water, enzymes and microorganisms.

TRANSFORMATION OF SIDE CHAINS

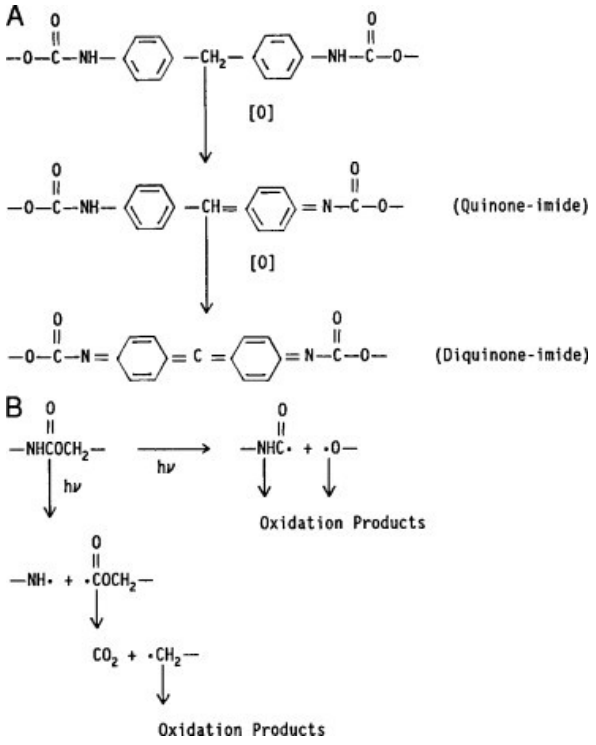


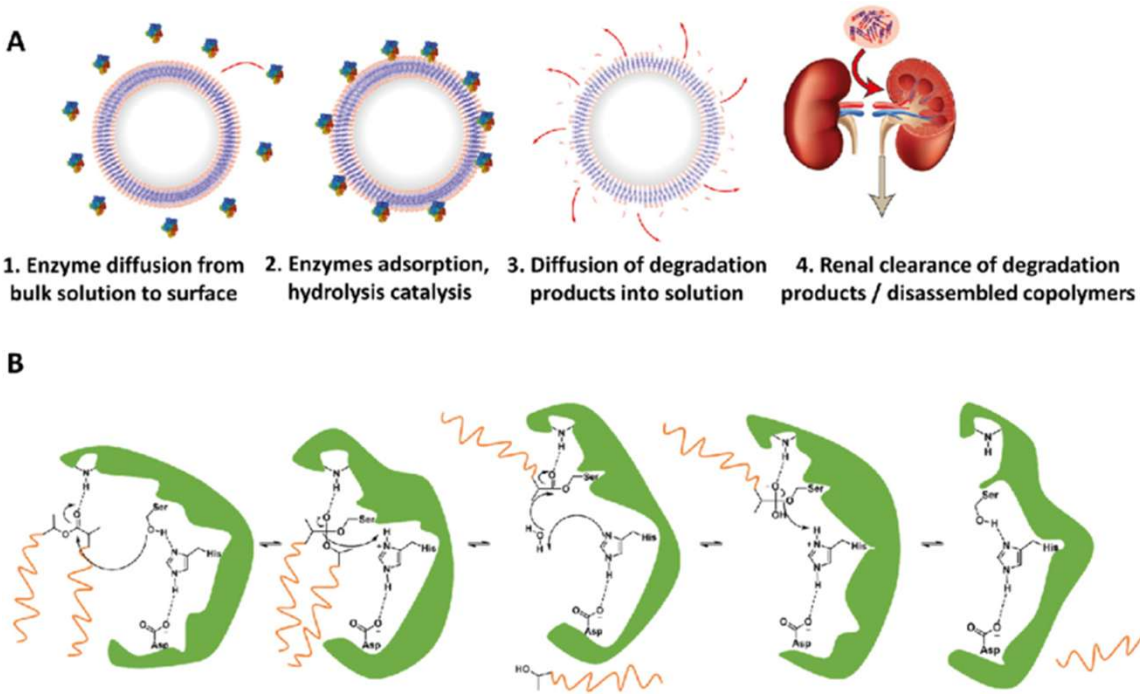
CLEAVAGE OF CROSSLINK



CLEAVAGE OF BACKBONE

ENZYMATIC DEGRADATION OF POLYMERS





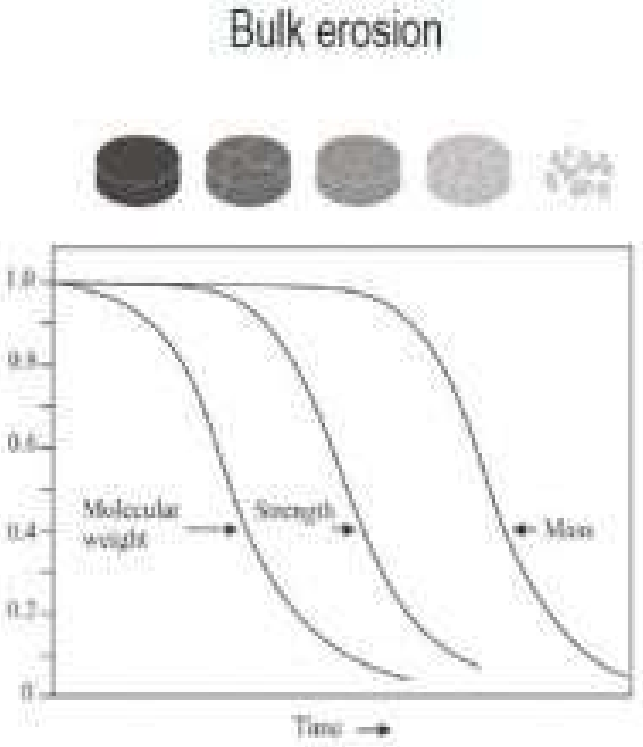
TYPES OF BIOEROSION

Bulk erosion

Happens throughout the sample

Ingress of water faster than the rate of degradation

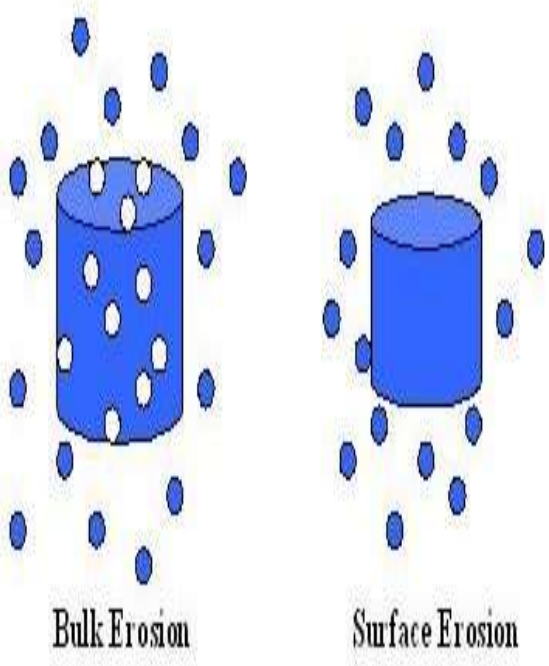
Ex: Polylactic acid (PLA)



TYPES OF BIOEROSION - CONT

Surface erosion

- Sample eroded from the surface
- Mass loss is faster than the ingress of water in the bulk
- Ex: Polyanhydrides



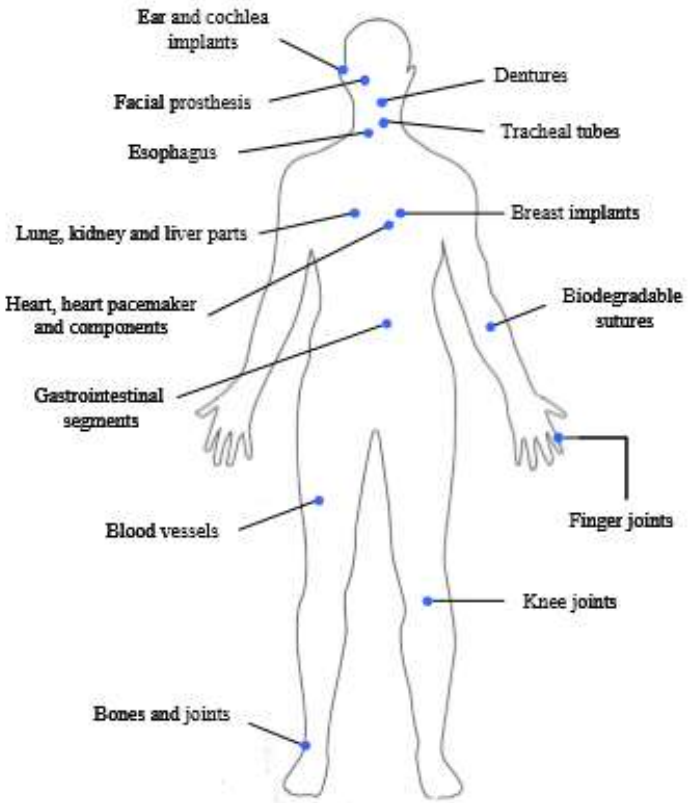
MECHANISM OF BIODEGRADABLE SCAFFOLDS

Polymers name	Degradation method	Application
Alginate	Enzymatic	Bone and cartilage tissue substitutes
Gelatin	Hydrolysis, dissolving, transformation, and enzyme-catalyzed decomposition	Cartilage cells
Starch/PVA	Hydrolytic	Bone tissue engineering
Collagen/PLLA	Enzymatic	Tissue engineering
PCL	Hydrolytic (surface erosion)	Drug delivery and tissue engineering
PGA	Hydrolytic	Tissue-engineered vascular grafts
PLA/thermoplastic polyurethane	Enzymatic	Tissue engineering

MSB Reddy et al., 2021

APPLICATIONS

WHERE CAN YOU FIND USES FOR POLYMERS?



Ear and cochlea implant	: acrylic, polyethylene, silicone, poly(vinyl chloride) (PVC)
Dentures	: acrylic, ultrahigh molecular weight polyethylene (UHMWPE), epoxy
Facial prosthesis	: acrylic, silicone, polyurethane
Tracheal tubes	: acrylic, silicone, nylon
Heart and components	: polyester, silicone, PVC, polyurethane, polyethylene, acetal
Lung, kidney and liver parts	: polyester, polyaldehyde, PVC
Eshophagus segments	: polyethylene, polypropylene (PP), PVC
Blood vessels	: PVC, polyester, polyurethane
Biodegradable sutures	: polyurethane
Gastrointestinal segments	: silicone, PVC, nylon
Finger joints	: silicone, UHMWPE
Bones and joints	: acrylic, nylon, silicone, polyurethane, PP, UHMWPE
Knee joints	: polyethylene

SOME POLYMERS THAT ARE USED İN BİOMEDİCAL APPLICATIONS

Polyethylene
PEEK (Polyether ether ketone)
Polyacrylic acid (PAAc)
Polyvinyl Chloride (PVC)
Polyethylene Terephthalate (PET)
Polyurethane (PU)
Polycarbonate (PC)
Poly(methyl methacrylate) (PMMA)
Polyethylene Glycol (PEG)
Poly(glycolic acid) (PGA)
Poly(lactic-co-glycolic acid) (PLGA)
Polydioxanone (PDO)
Polyhydroxyalkanoates (PHA)
Silicone Rubber
Hydrogels: poly(acrylamide)
Chitosan (natural bşopolymer)
Etc.

The choice of polymer for a specific application is dictated by multiple factors such as application-specific requirements, including:

- Biocompatibility
- Bioinertness
- Biodegradability
- Durability
- Optical Properties
- Processing Requirements
- Surface Properties
- Physical Properties (strength, elastic modulus, etc.)
- Regulatory Compliance
- Economic Considerations

You don’t have to memorise those every word,
but you have to have a sense of understranding that some specific application require materials with specific properties such as....

Medical Packaging

The requirements for medical products packaging are medium and high barrier protection
The demand for sterilizable packaging for hospital and industry is being met by special paper, plastic films, and other nonwoven materials.
Examples of medical packaging:
chlorinated tetrafluoroethylene (CTFE) is used in an antibiotic test kit where even trace amounts of moisture would violate the accurace of the test,
poly(ethylene terephthalate) (PET) is used as medical containers, the advantages over glass include no breakage of filling lines and higher output.



Medical Packaging





Blister pack for pharmaceutical tablets / pills – usually made of PVC

The principal advantages of PVC are the low cost and the ease of thermoforming. The main disadvantages are the poor barrier against moisture ingress and oxygen ingress; moreover PVC has a negative environmental connotation due to its chlorine content. In the case of blister packaging the PVC sheet does not contain any plasticizer and is sometimes referred to as Rigid PVC or RPVC. In the absence of plasticizers, PVC blisters offer structural rigidity and physical protection for the pharmaceutical dosage form. On the other hand, the blister cavity must remain accessible by the push-through effect and the formed web may not be too hard to collapse when pressed upon; for this reason the PVC sheet thickness is typically chosen between 200µ to 300µ depending on the cavity size and shape. Most PVC sheets for pharmaceutical blisters are 250µ or 0.250 mm in thickness.

Examples

RESIN	PRIMARY FEATURES	TYPICAL APPLICATIONS
Polyethylene	Processing ease, chemical resistance	Caps, needle hubs, medical packaging, waste bags
Polypropylene	Autoclavibility, contact clarity	Syringes, specimen collection cups
Gen-purpose polystyrene	Transparency	Petri dishes, labware, test tubes
High-impact polystyrene	Toughness, opacity	Home test kits, diagnostic equipment, housing
Styrene-acrylonitrile	Chemical resistance, transparency, toughness	Diagnostic components, fluid handling devices, flat plate dialyzers
Acrylic	High light-transmission rates, chemical resistance	I.V components, specimen-collection containers
Acrylonitrile-butadiene-styrene	Toughness, low and high gloss, good processability	Home test kits, housing, surgical staplers, I.V.connectors

Examples

Polycarbonate	Chemical and heat rasistance, toughness, transparency	Blood centrifuge bowls, cardiotomy reservoirs, profusion devices, hemodialyzers
Polyester	Chemical resistance	I.V.components, catheters, surgical instrument, housings
Poly(vinyl chloride)	Transparency, good scuff resistannce	Blood bags, catheters, cannulae, corrugated tubing, renal care product, transfusion supplies, face masks
Polyurethane	Good chemical resistance, toughness, good processability	Catheters, tubing, I.V.connectors, drug delivery systems
Polyetherimide	Autoclavibility, chemical resistance	Sterilization trays
Polysulfone	High heat resistance	Medical trays

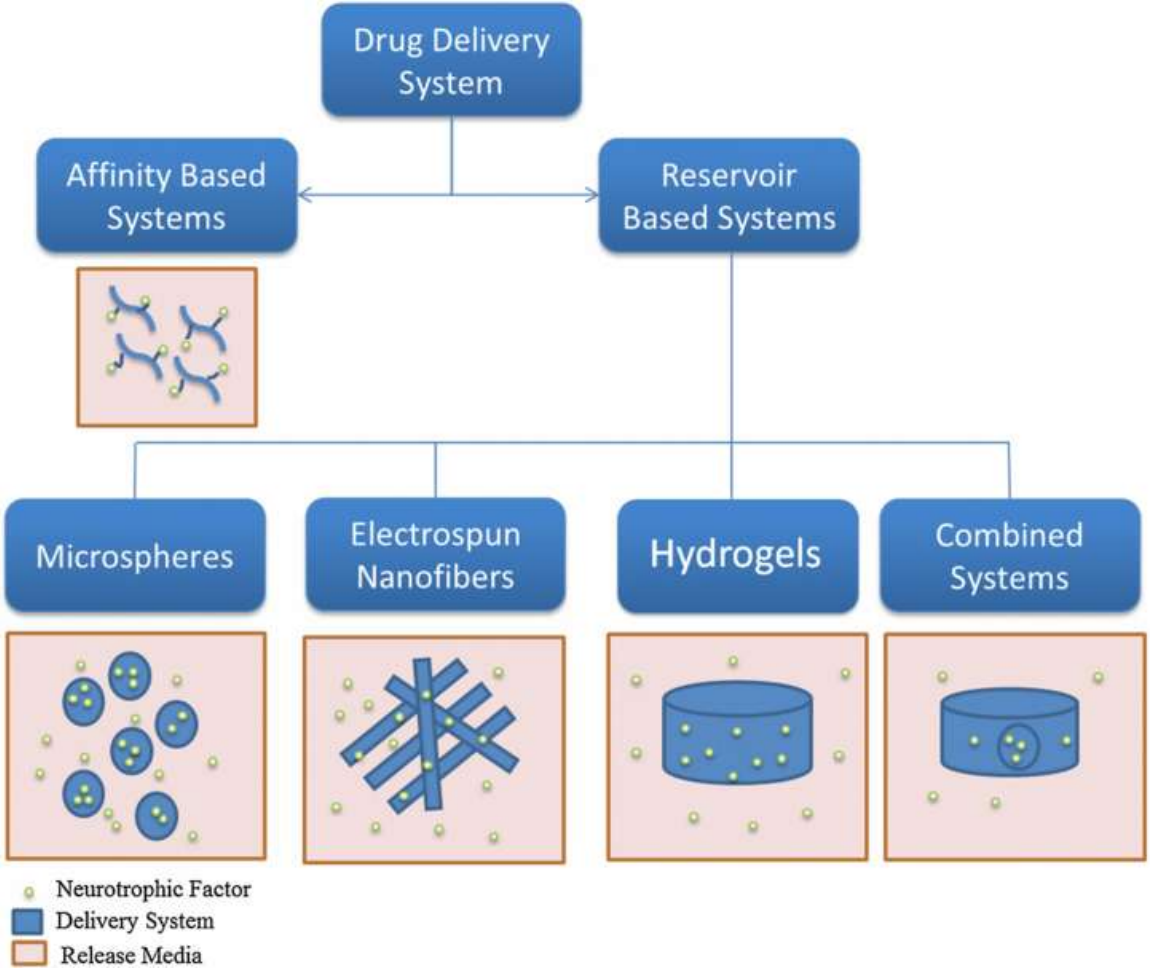
POLYMERS IN DRUG DELIVERY

Drug delivery refers to approaches, formulations, technologies, and systems for **transporting a pharmaceutical compound in the body** as needed to safely achieve its desired **therapeutic effect**.

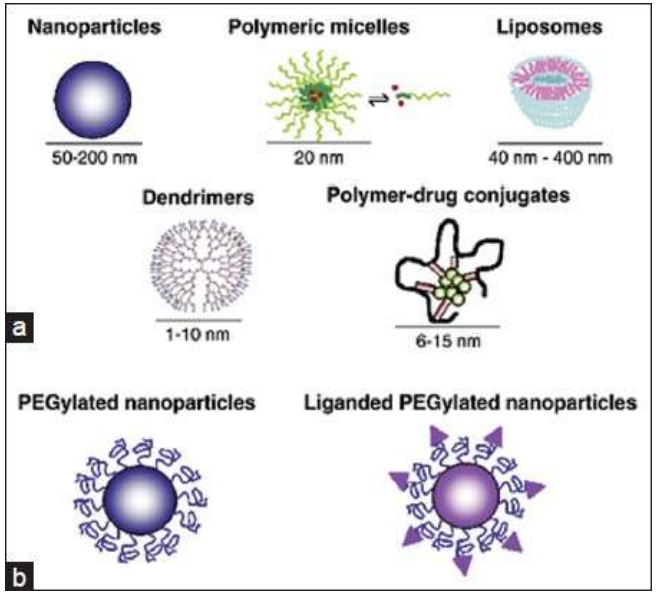
Ideal requirements of implantable drug delivery systems:

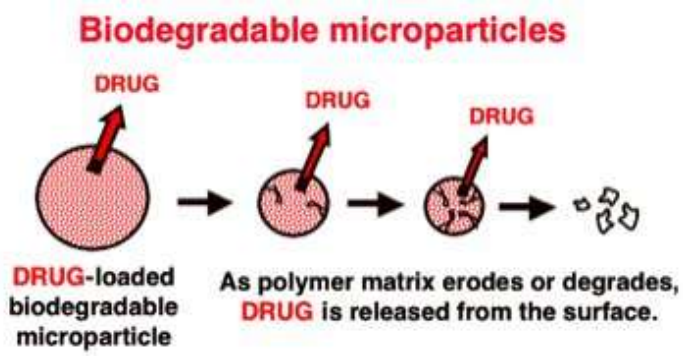
- Environmentally stable.
- Biocompatible.
- Sterile.
- Biostable.
- Improve patient compliance by reducing the frequency of drug administration over the entire period of treatment.
- Release the drug in a rate-controlled manner that leads to enhanced effectiveness and reduction in side effects.
- Readily retrievable by medical personnel to terminate medication.
- Easy to manufacture and relatively inexpensive.

POLYMERS IN DRUG DELIVERY



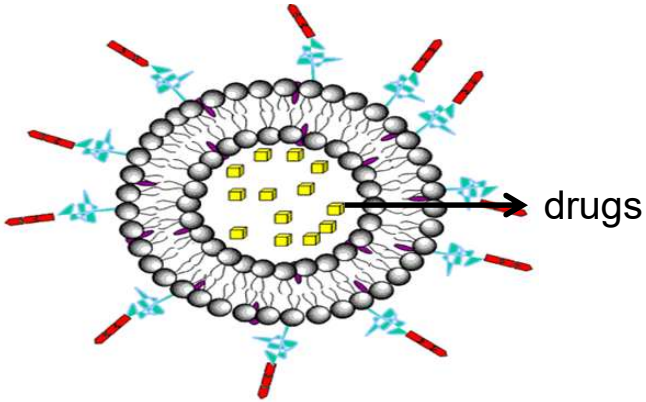
POLYMERS IN DRUG DELIVERY





Carriers which are able to biodegrade include:

- ✓Liposomes;
- ✓Microspheres made of the biodegradable polymer poly(lactic-co-glycolic) acid
- ✓albumin microspheres;
- ✓synthetic polymers (soluble);
- ✓nanofibers
- ✓Protein-DNA complexes;
- ✓protein conjugates;
- ✓erythrocytes
- ✓virosomes
- ✓Dendrimers - is hot in biomedical research field.



POLYMERS IN DRUG DELIVERY

Pill coatings
Rapid dissolving capsules
- Fluid driven
- pH driven

Degradable delivery vesicles

Benefits of Tablet Coating

Tablets coating mask the taste, odour, or colour of the drug. Tablets coating control the release of the drug from the tablet. It provides physical and chemical protection and protects the drug from the gastric environment of the stomach (acid resistant enteric coating). Incorporate of another drug or formula adjuvant in the coating to avoid chemical incompatibilities or to provide sequential drug release, improvement of pharmaceutical elegance by use of special colours and contrasting printing can also be obtained from tablet coating.

- **PLGA (poly lactic-co-glycolic acid)**: Controlled release microspheres.
- **Poly(N-isopropylacrylamide) (PNIPAM)**: Thermoresponsive hydrogels.
- **Chitosan**: Mucoadhesive and antimicrobial properties.
- **Ethylcellulose**: Used in controlled oral delivery systems.

Advancements

- **Smart polymers** that release drugs in response to pH or temperature.
- **Microneedle patches** using PCL or chitosan for transdermal drug delivery.
- FDA-approved products like **Lupron Depot®** and **Risperdal Consta®** use PLGA-based microspheres for sustained release.



DISSOLVING CAPSULES



The two main types of capsules are:
1. Hard-shelled capsules, which are typically made using gelatin and contain dry, powdered ingredients or miniature pellets made by, e.g. processes of extrusion or spheronisation.
2. Soft-shelled capsules, primarily used for oils and for active ingredients that are dissolved or suspended in oil.



Both of these classes of capsules are made from aqueous solutions of gelling agents like:
✓Animal protein mainly gelatin;
✓Plant polysaccharides or their derivatives like carrageenans and modified forms of starch and cellulose.
Other ingredients can be added to the gelling agent solution like plasticizers such as glycerin and/or sorbitol to decrease the capsule's hardness, coloring agents, preservatives, disintegrants, lubricants and surface treatment.

BIODEGRADABLE POLYMERS AS DRUG CARRIERS

Poly(amides)

- Natural Polymers
  - Remain attractive because they are natural products of living organism, readily available, relatively inexpensive, etc.
  - Mostly focused on the use of proteins such as gelatin, collagen, and albumin

Polyesters

- Lactide/Glycolide Copolymers
  - Have been used for the delivery of steriods, anticancer agent, antibiotics, etc.
  - Poly-L-lactide (PLLA) is found as an excellent biomaterials and safe for in vivo (Lactic acid contains an asymmetric α-carbon atom with three different isomers as D-, L- and DL-lactic acid)
  - Poly(lactic-co-glycolic acid) (PLGA) is most widely investigated biodegradable polymers for drug delivery.
  - Lactide/glycolide copolymers have been subjected to extensive animal and human trials without any significant harmful side effects

Polymer Processing

- Drug-incorporated matrices can be formulated either compression or injection molding
- Polymer & drug can be ground in a Micro Mill, sieve into particle size of 90-120 μm, then press into circular disc
- Alternatively drug can be mixed into molten polymer to form small chips, then it is fed into injection molder to mold into desired shape

HEART VALVES

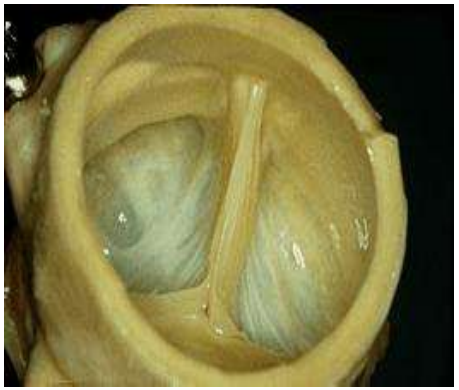
Damaged heart valves, weakened arterial walls, and blocked arteries constitute some of the commonest cardiovascular disorders.

Silicone rubber is used because of its inertness, elasticity, and low capacity to cause blood clotting.

Poly(tetrafluoroethylene)

SUBSTITUTE HEART VALVES





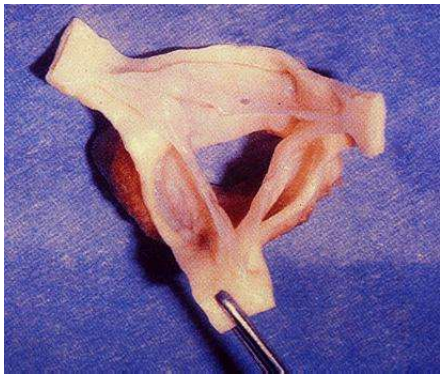




Figure 4.3 This artificial heart valve is coated with Silizone, a biocompatible material that allows the body to accept the implant. (from http://www.sjm.com/devices/).

BONES, JOINTS, AND TEETH

- Occasionally repaired with the use of polyurethanes, epoxy resins, and rapid curing vinyl resins.
- Silicone rubber rods and closed cell sponges- replacement finger and wrist joints.
- Elbow joints- vinyl polymers and nylon
- Knee joints- cellophane and, more recently, silicone rubber
- Poly(methyl methacrylate) is the principal polymer used both for acrylic teeth and for the base material

BIOMEDICAL APPLICATIONS: ARTIFICIAL HIP JOINTS

UHMWPE: acetabular caps in hip implants and patellar surface of knee joints

HDPE used as pharmaceutical bottles, fabrics

Others used as bags, pouches, tubes etc.





Contact Lenses And Intraocular Lenses

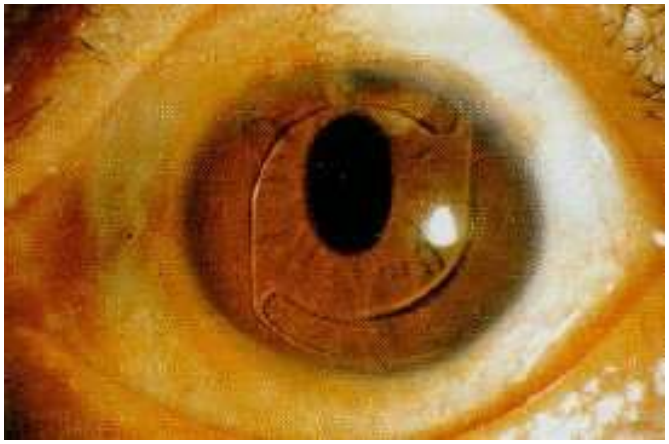


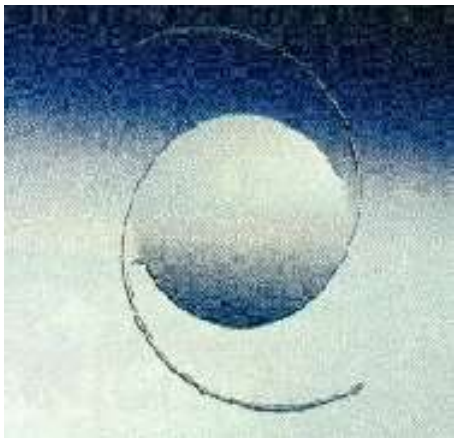
INTRAOCULAR LENS

3 basic materials - PMMA, acrylic, silicone

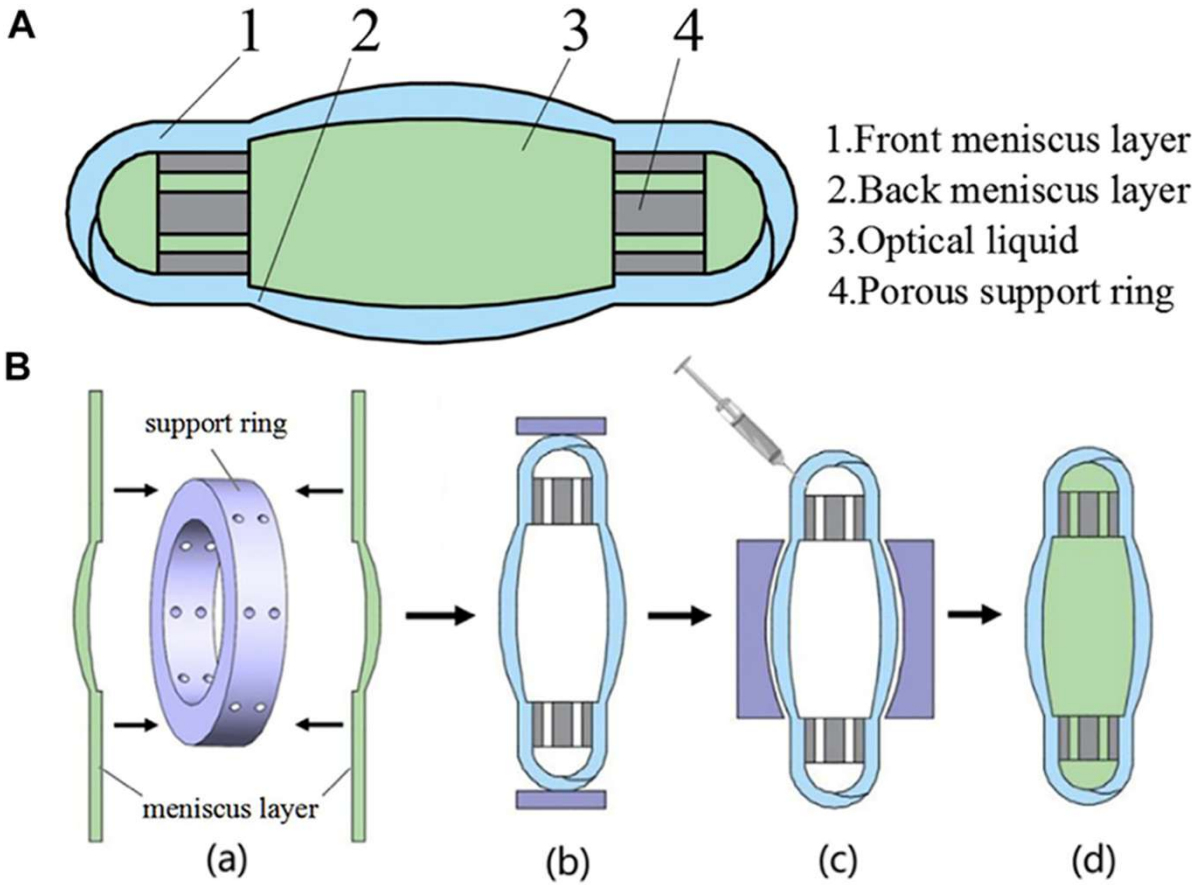
Polymethylmethacrylate (PMMA, lucite, acrylic, plexiglas)

- acrylics
- transparency
- tough
- biocompatible





POLYMERS IN OPHTHALMIC APPLICATIONS



Types of Polymers Used

- **Silicone**: Used in intraocular lenses (IOLs).
- **Hydroxyethyl methacrylate (HEMA)**: Common in contact lenses.
- **PEG-based hydrogels**: Drug-eluting ocular inserts.
- **Collagen and hyaluronic acid**: For corneal wound healing.

Advancements

- **Drug-eluting contact lenses** to treat glaucoma.
- **Thermo-responsive in situ gels** for sustained ocular drug delivery.
- **Bioprinted corneal scaffolds** using collagen-methacrylate blends.

SILICONES

Catheters
Drainage tubes
Contact lenses
Medical adhesives
Coating for pacemakers





Artificial Kidney And Hemodialysis Materials

- The function of a kidney is to remove low molecular weight waste products from the bloodstream.
- Artificial kidneys have function by passage of the blood between the walls of a dialysis cell which is immersed in a circulating fluid.
- Cellophane- Semipermeable dialysis membranes
- The polymer is "heparinized" to prevent blood clotting-polycarbonate or cellulose acetate fibers.

Oxygen-Transport Membranes

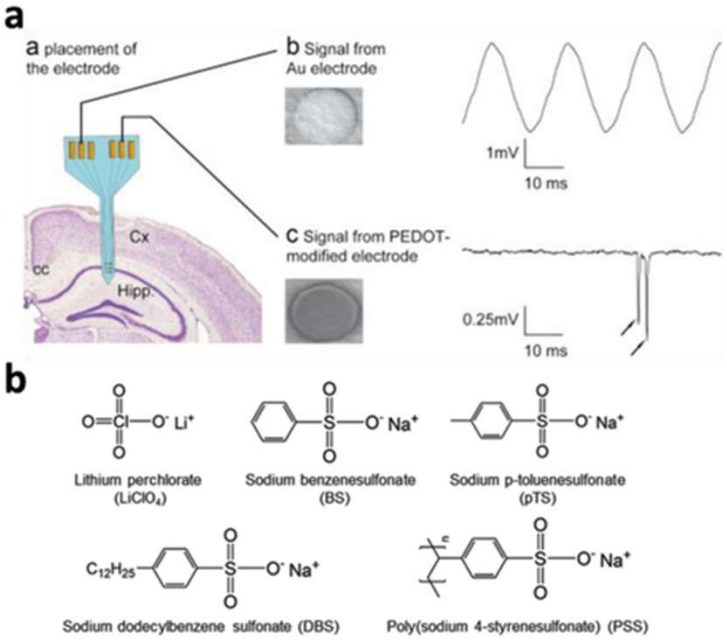
- ✓Surgical work on the heart frequently requires the use of a heart lung machine to circulate and oxygenate the blood.
- ✓Poly(dimethylsiloxane) membranes are highly efficient gas transporters.
- ✓It is of interest that silicons rubber has approximately six times the oxygen permeability of fluorosilicones.

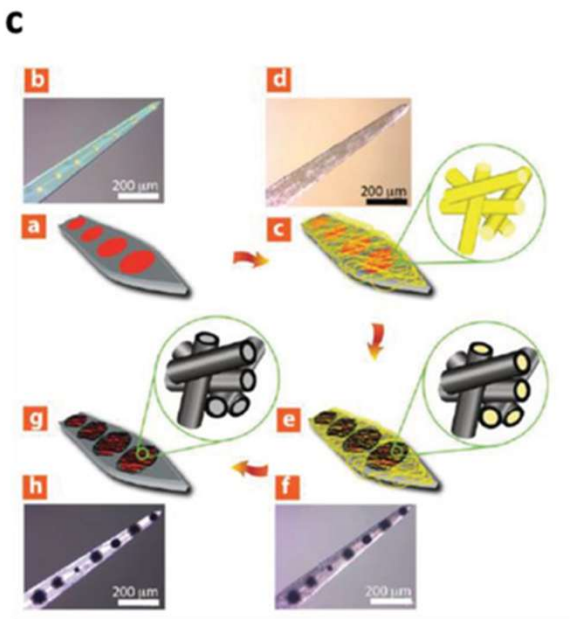
SURGICAL SUTURES

- Poly(glycolic acid), or condensation copolymers of glycolic acid with lactic acid.
- A high tensile strength and is compatible
- The polymer degrades by hydrolysis to nontoxic glycolic acid.



POLYMERS IN NEURAL IMPLANTS





Types of Polymers Used

- **Polypyrrole (PPy)** and **PEDOT:PSS**: Electrically conductive polymers.
- **Polyimide**: Flexible substrates for microelectrodes.
- **PEG and PCL**: Biodegradable carriers for nerve guidance conduits.

Advancements

- **Bioresorbable electronics** for temporary stimulation or monitoring (Rogers et al., 2016).
- **Conductive hydrogels** for spinal cord injury repair.
- **3D printed nerve conduits** combining PCL with Schwann cells.

POLYMERS: BIOMEDICAL APPLICATIONS

- Polyamides (PA, nylon)
  - high degree of crystallinity
  - interchain hydrogen bonds provide superior mechanical strength (Kevlar fibers stronger than metals)
  - plasticized by water, not good in physiological environment

Used as sutures

- Polyvinylchloride (PVC) (monomer residue must be very low)
  - Cl side chains
  - amorphous, hard and brittle due to Cl
  - metallic additives prevent thermal degradation

Used as blood and solution bags, packaging, IV sets, dialysis devices, catheter, bottles, cannulae

POLYMERS: BIOMEDICAL APPLICATIONS

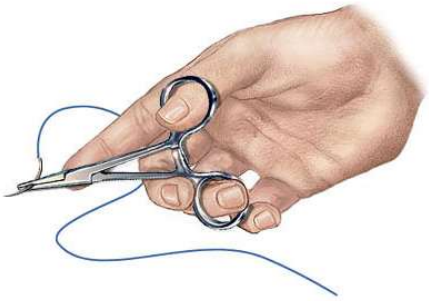
- Polypropylene (PP)
  - properties similar to HDPE
  - good fatigue resistance

Used as syringes, oxygenator membranes, sutures, fabrics, vascular grafts



- Polyesters (PET)
  - hydrophobic (beverage container PET)
  - molded into complex shapes

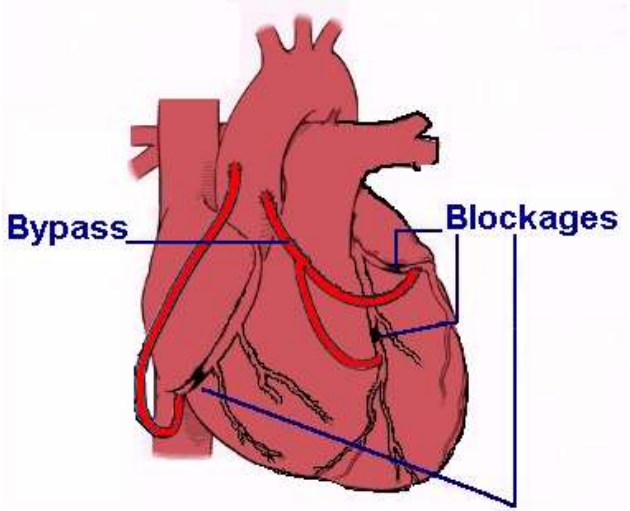
Used as vascular grafts, sutures, heart valves, catheter housings



POLYMERS: BIOMEDICAL APPLICATIONS

- Polytetrafluoroethylene (PTFE, teflon)
  - low coefficient of friction (low interfacial forces between its surface and another material)
  - very low surface energy
  - high crystallinity
  - low modulus and strength
  - difficult to process

catheters, artificial vascular grafts





Synthetic vascular grafts from W.L.Gore

POLYMERS: BIOMEDICAL APPLICATIONS

- Rubbers
  - latex, silicone
  - good biocompatibility

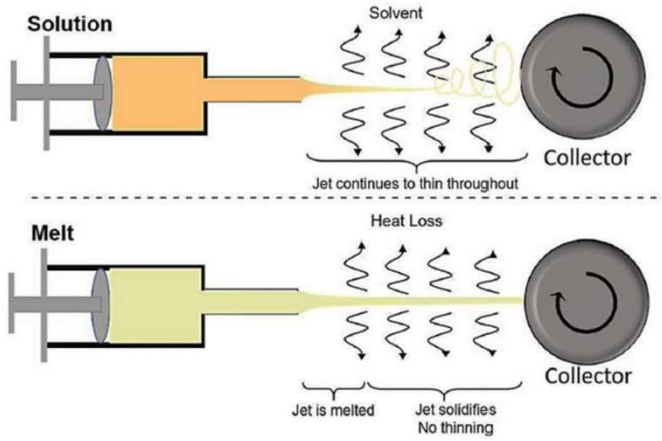
Used as maxillofacial prosthetics

- Polyurethanes
  - block copolymer structure
  - good mechanical properties
  - good biocompatibility

tubing, vascular grafts, pacemaker lead insulation, heart assist balloon pumps

LATEST TRENDS IN BIOMEDICAL APPLICATIONS

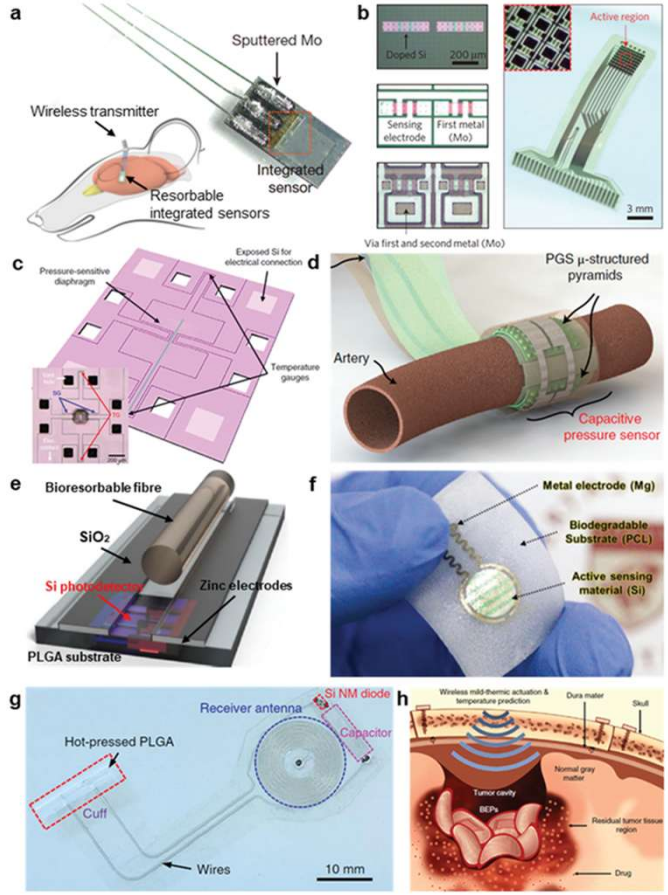
- **3D Printing/Additive Manufacturing**: Used to fabricate patient-specific implants with intricate architectures for bone and dental applications. For instance, 3D-printed PCL scaffolds seeded with mesenchymal stem cells show promising results in cranial defect repair (Bose et al., 2013).
- **Electrospinning**: Enables production of nanofiber meshes with high surface-area-to-volume ratio, ideal for skin, vascular, and neural tissue engineering.
- **Smart Polymers**: Shape-memory polymers are being used in minimally invasive surgery. Thermo-responsive polymers release drugs at fever temperatures, enabling targeted therapy.



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LATEST TRENDS IN BIOMEDICAL APPLICATIONS

- **Biohybrid Materials**: Combining polymers with collagen or fibrin to create hybrid scaffolds with mechanical and biological integration.
- **Nanocomposites**: Integration of silver, graphene oxide, or nano-hydroxyapatite enhances mechanical strength, antimicrobial activity, and osteogenesis.
- **Bioresorbable Electronics**: Implanted devices such as sensors or stimulators that degrade after performing their function, avoiding removal surgery (Rogers et al., 2016).



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